



INTERVIEW Q&A GUIDE

SALESANDTRADING.ORG
NEW YORK, NEW YORK



Stay Up-To-Date with Markets via The Market Making Newsletter

As you can hopefully tell from going through these guides, I love writing about markets and trying to break down complex topics as simply as possible.

So, I've thought quite a bit about how I could write more regularly about markets and keep you updated on how those on the buy-side and sell-side are thinking about things.

In the end, I decided to create a weekly newsletter – delivered to your inbox every Sunday evening – that dives deep into what themes and stories are currently moving markets.

My aim with the newsletter is to share the institutional sales and trading perspective and walk you through some of the sell-side research that has been most discussed on trading desks across asset classes over the past week.

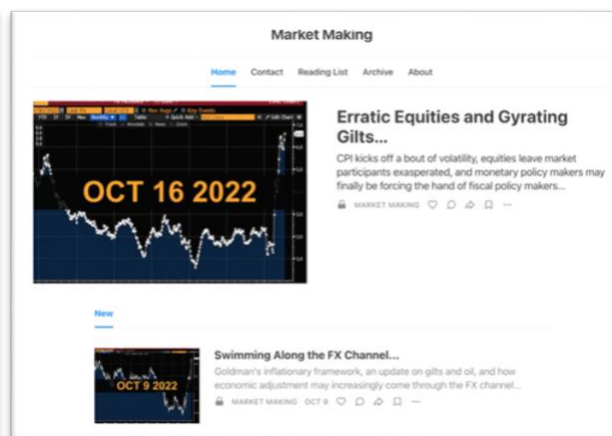
Each Sunday's newsletter is around 5,000-7,000 words and will probably take you 25-35 minutes to read. So, the newsletters aren't exactly short (brevity isn't my strong suit!).

But if you're deeply interested in markets and enjoy my writing style, then I'm sure you'll love reading the Market Making newsletter each week.

Given how much time and effort goes into creating the newsletter each week, I do charge a small amount for full access to it. However, the first part of each week's newsletter will always be completely free as I want everyone to be able to enjoy and benefit from it.

You can read past newsletter editions and subscribe here: MarketMaking.Substack.com

Creating the newsletter each week has been a lot of fun and I hope reading it becomes something you look forward to doing every Sunday evening!

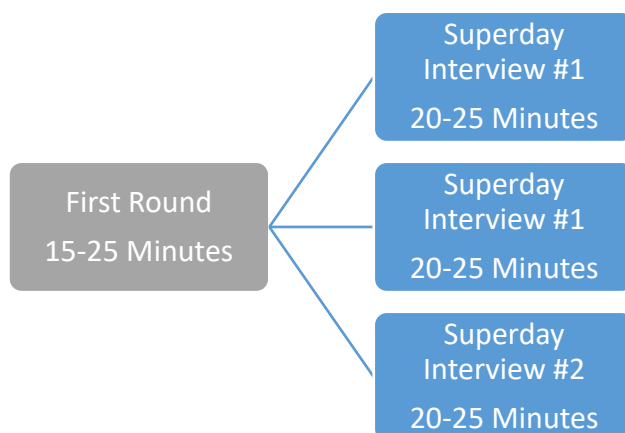




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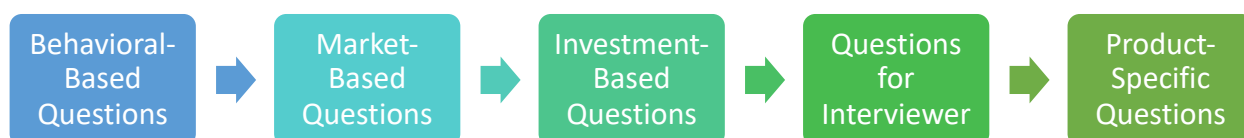
As mentioned in the Overview Guide, the S&T interview process at almost every large bank will involve one first round interview via Hirevue or phone screen.

This will be followed by a superday, which is composed of three 20-25-minute interviews with S&T professionals in almost invariably a one-on-one setting.



Types of Questions

The best way to think about S&T interview questions is through five categories.



▪ Behavioral-Based Questions

These are the questions you're guaranteed to get in every S&T interview. They always start off an interview and can comprise either almost the entire interview or the first 5-10 minutes of it.

These are the questions that are most open-ended and require the most thought – not just memorization – to succeed in answering.

In S&T the “behavioral” bucket of questions is reasonably large; covering questions of personal interests, ethics, and how you think about S&T and keep up to date on markets.



▪ **Market-Based Questions**

No one expects you to be able to rattle off your own research report on a certain stock or expound on the general macro-economic environment at length.

However, following the markets is (obviously!) a good sign you'll enjoy being in S&T and so you should expect at least some market-based questions thrown your way that will focus on where markets are now and how they operate.

▪ **Investment-Based Questions**

Once again, no one expects you to have an airtight thesis for an investment idea or to be able to talk at length about asset allocation.

What investment-based questions are meant to do is show your capacity to clearly and concisely communicate something and persuade the interviewer (within reason) that you've thought it through.

The questions I give in this section are the standard ones; don't expect anyone to ask you to pitch a certain MBS trade (unless you specifically say you love mortgage backed securities and follow them closely, which I don't advise you saying!).

▪ **Questions for Interviewer**

These are the single most underrated form of questions. At the end of every S&T interview, five minutes or so will be left for you to ask your interviewer questions.

Thoughtful questions – that I provide examples of – are a critical way not only to stand out in an interview, but also leave a great final impression on the interviewer.

You will have the opportunity to ask your interviewer questions. I can't state enough how critical it is to ask good ones.

▪ **Product-Specific Questions**

Product-specific questions involve you showing some *basic* knowledge of certain types of S&T product (rates, CLOs, MBSs, etc.). These questions will normally only crop up if you express an interest in certain products, which will then lead the interviewer to throw you some basic questions to gauge if you know what you're talking about.

Again, no one is expecting you to be an expert. However, product-specific questions will come your way if you express an interest in a specific product so you should not be caught flat footed on these.

The questions in this guide are the most basic, fundamental ones from the Desk Guides, which you should review for a deeper level of understanding.



First Round vs. Superday Questions

The first-round interview will take one of two fundamental forms: either a recorded video format (Hirevue) or a phone interview.

If the first round is some recorded video format – with pre-set questions – then it will likely be viewed by someone within HR or potentially someone within HR and a S&T professional. They will then decide if you get a superday.

If the first round is done via a phone call – as is still common at some firms – then it will be directly with a S&T professional and they alone will decide if you get a superday.

In either case, the composition of questions will be roughly as follows:



As you can see, the first round is dominated by behavioral-based questions, which can be a bit more involved and slightly technical than behavioral questions within the traditional investment banking context (as you'll soon see).

For HireVue interviews, there is a bank of hundreds of questions that you randomly will be assigned. However, all of them will be more or less permutations of those listed in this guide.

For example, perhaps instead of being asked about a markets-related article you've read recently, they'll ask you about a certain trend in the markets you've been following. Obviously, these are different questions, but the answer to both questions can be the same!

So as long as you prepare thoroughly there should be nothing that catches you off guard in a HireVue. Also, you'll be getting 5-7 questions (depending on the firm) in a first-round HireVue so not knocking one of them out of the park is perfectly fine.

For superdays you'll be either interviewing via a live video stream or in person with S&T professionals. The interview styles will thus vary significantly depending on who the S&T professionals are. An equity derivatives trader may want to ask more market-based questions and folks in investment grade sales may want to ask primarily behavioral questions.

For this reason, it's important to be as broadly prepared as possible and to follow my interview "do's and don't's" in order to make sure you don't set yourself up for failure.



Generally speaking, you can expect the questions asked to be broken down as follows:



A Note on Brainteasers

I've been asked why I haven't included any brainteasers in this course. There are four primary reasons:

1. Brainteasers were quite common – from what I've heard – 10-15 years ago but are hardly used at all nowadays; I personally never got one in four S&T superdays
2. Brainteasers are hard to prepare for (that's the reason why they're sometimes asked!) and so it's far better to spend your time on things that are much, much more likely to be asked and you can better prepare for
3. If you do get a brainteaser, getting it right or wrong will not have much weight. In other words, if you try to answer the brainteaser well, but get it wrong, that won't result in you *not* getting the offer
4. On the off chance you get brainteasers, it will just be one or two questions at most and make up just a few minutes of your interview. It's far better to knock out the behavioral questions than it is to get the brainteasers right

Ultimately, I created this entire course to try to provide you the most comprehensive preparation possible without needing you to spend hundreds of hours doing prep.

Studying brainteasers will at best *slightly* increase your odds of getting an offer and at worst will be a waste of time that distracts you from preparing for the questions you actually *will* get.

Interview Do's and Don't's

In the Overview Guide I spent a significant amount of time going over how you should prepare for interviews generally and what will make you stand out. You should make sure to go and read that if you haven't already.

Below are some do's and don't's for answering the actual interview questions:



DO keep all your answers below two minutes

Some of the interview questions you face will be broad and some people are prone to rambling (myself included!). Make sure to keep in mind that your total interview is just 20-25 minutes.

If you're spending more than two minutes answering a question, you're spending potentially over 10% of your entire interview on one question! As a rule of thumb, you should never spend more than two minutes on any one question.

DO be humble

As I've said many times throughout this course, the one character trait you can't have in an interview is arrogance. You should always aim to be self-deprecating where possible.

DON'T say you know exactly what desk you want to be on

It's a good idea to have a general idea of what area of S&T you want to be in; especially if the program you're applying to is non-rotational as some are.

However, you should never give *one* specific desk that you want to be placed on. This is because folks will view it as being quite arrogant given that you haven't actually spent time in S&T yet *and* that desk may or may not have places open. You should say a broad area (credit) or give a few different desks.

DON'T try to answer what you can't

If you get tossed a difficult question about markets or about specific products and how they operate, don't try to answer if you have absolutely no idea. It's far better to be honest in saying you aren't sure and (if you can) give an answer.

For example, you would caveat your answer by saying, "Well, I'm really not sure about this one, but if I had to give my best answer it would be..."

DO ask at least two questions to your interviewer

The majority of interviewees do not come prepared with questions to ask their interviewer at the end. This is one area where you can really shine and leave a positive impression at the end of the interview.

You should ask at least two questions. Be mindful of time and if you still have some left you can ask another one. Three should be the top, however.



DO thank your interviewer at the end

When the interview ends you should always thank your interviewer for taking the time. All you need to say is, “Thanks for taking the time”. No need for anything else.

DO come prepared with copies of your CV and cover letter

You should make sure to always bring a few copies of your CV and cover letter to your interview just in case they don’t have it. They obviously should, but it shows preparedness on your side. To be safe, you should have five copies of each.

DO bring a pen and paper

It’s rare you’ll have to actually use it, but once again you want to be prepared if they ask you to draw the shape of the yield curve or something like that. So, what you should bring with you to a superday is a pen, paper, and copies of your CV and cover letter.

DON’T talk about high school accomplishments (unless they’re exceptional)

As recruiting timelines have been brought forward significantly, many interviewees now talk about things they did in high school (perhaps because they’ve only been in college two years or less).

Unless you were an exceptional athlete, musician, or won nation-wide academic awards avoid talking about high school accomplishments.

Behavioral-Based Questions

Walk me through your resume?

This is almost always the first question you get. Obviously, every answer will be unique – since it’s your own background! – but here are some rules to follow:

- Remember to keep your answer under two minutes
- You can make mention to where you were born and raised, but you should not spend time talking about anything you did prior to college
- You should spend no more than 30-60 seconds on each major resume line item (e.g. each prior internship you have or major involvement at college)
- You should try to ensure – like in your cover letter – that you’re telling a story that makes it obvious why you’re interested in S&T (even if you don’t explicitly explain why in this answer)

This is a question asked nearly 100% of the time. You should type out an answer and read it over while timing yourself to make sure you’re including everything you want to. Just



make sure when you actually answer this question in the interview it doesn't sound entirely scripted!

"Sure, so I was born and raised in Tampa Bay and ended up deciding to go to Dartmouth since it seemed like it would be an entirely different world than what I was used to growing up and it definitely has been. At Dartmouth I'm studying economics - which is a bit typical for someone trying to get into finance, I know - but I really do enjoy my macroeconomic policy classes.

Last summer I was at [x] doing an asset management internship. I really enjoyed the firm and the people, and it gave me the chance to get to read lots of research reports from places like [bank you're interviewing at]. I began to realize at [x] that I wanted to feel a bit more connected to the markets and get a bit more specialized in an actual product, which obviously S&T provides the capacity to do.

Of course, I'd never presume to know exactly what desk I'd like to be on prior to even beginning. But, like I said, I have enjoyed my macroeconomic classes where we talk quite a bit about central banking policy, and I really enjoyed reading rates research last summer, so I think somewhere in rates may be a good fit. But again, I'm sure I will probably entirely change my mind if I'm fortunate enough to get an offer. "

Why S&T?

Some people try to develop a very personal and unique answer here. I think giving a standard answer, with maybe a bit of your personal touch on it, is far better.

You should not say that you have followed the markets since you were a child or that you love investing on your own. That doesn't clearly communicate that you know what S&T is (it's not picking stocks on an app). Were you following global macro trends as a teen? Are you doing interest rate swaps in your spare time? Obviously, the answer is likely no.

Your why S&T should clearly communicate you know what S&T is and that you are equipped to succeed in it.

"I think S&T provides a great opportunity for someone who is intellectually curious and has a mix of quantitative and qualitative attributes. To me, it's an incredible chance to get a front row seat to how a given market – depending on what desk you're on – operates and to see how clients think about a given market at any given time.

I've talked to those within S&T at various levels, so I know it's not all roses, of course. But I still think on a risk-adjusted basis there's hardly a better job to have out of undergrad than in S&T given my personality and interests."



What has been your impression of [bank] after talking to some of our S&T folks?

"I've really enjoyed talking to folks here. I think it's obvious that they're not only very smart - which is probably necessary to get an offer, I suppose - but also that they're very generous with their time and seem to genuinely love their jobs."

What do we even do in S&T?

"Well, obviously S&T is kind of a catchall term that could denote desks that are in equities, commodities, fixed income, or FX. Further, S&T encompasses desks that are more flow-oriented and those that are more illiquid or structured and roles that range from managing a book to dealing with clients to structuring novel transactions."

What S&T really boils down to in 2020 - from what I understand - is serving clients by being a leading market maker and managing risk. This is accomplished by traders, sales people, salestraders, quants, and structurers (depending on the desk).

Like I said, obviously there's a huge diversity in roles and desks. But I think it all comes down to serving clients and managing risk."

What's market making? Do we take proprietary bets on assets here?

"After the great financial crisis, new regulations were put in place that severely crimped the capacity for most desks to operate in a fully proprietary manner via the Volcker rule."

As a result, most desks are oriented towards market-making which is providing clients with the capacity to execute trades of various volumes via giving either a bid, ask, or spread to them."

Obviously, desks don't have the capacity to fully hedge out all their exposure at any given time, so it's not that market-making really means just taking the spread on trades. Every desk will warehouse and manage some level of risk so most desks operate in that kind of middle ground between proprietary and market making."

What are the major divisions or product-groups of S&T?

"The major divisions within S&T are equities, commodities, fixed income, and FX. Obviously, there's a bit of nuance here as some desks are a bit cross-asset and obviously there's quite a diversity within each division; for example, fixed income would cover swaps, treasuries, money markets, and MBS."



Who are our clients?

"What all clients have in common is that they are very large and need to transact in size. This could mean you're dealing with fast money (hedge funds) or pension funds, endowments, private equity firms, or corporate clients. Who you deal with depends on the desk, of course. Treasuries will be bought by lots of large institutional buyers, whereas distressed debt will probably have a bend toward hedge funds."

Have you considered investment banking? Why not that?

"I think everyone interested in S&T has probably also considered investment banking. To be honest, from friends who have gone into investment banking, what they all say is that it can be quite boring and repetitive – despite the very long hours - and for the most part you feel disconnected from the market."

While obviously investment banking is a great job to have at any level, I'm just a bit too [extroverted, analytical] to think it would be a great fit for me personally."

What are your major concerns about S&T?

"Well, unlike in investment banking where the work is quite similar regardless of the specific coverage group you're in, in S&T obviously you have a wide variance and you always run the risk of getting onto a desk that you think is right for you, but turns out not to be."

Further, obviously there are regulatory and technological changes that have impacted S&T that one always has to weigh when thinking about where you want to start your career."

Ultimately, I've thought a lot about my decision to pursue S&T and while I obviously have imperfect information, the pros much more than outweigh the cons."

Is there a particular area or areas of S&T that interests you?

This is where you should state what areas of S&T you're interested in as I covered in the Overview guide. An important thing to keep in mind is that you want your answer to be broad both because you don't want to come across as arrogant (knowing exactly where you want to be) and you don't want very detailed product follow-up questions that may be hard to answer.

An example of a great answer is as follows:

"Obviously it's hard to really have a grasp on what a desk does, what the culture is like, etc. before you begin. However, with that said, I do find credit to be really interesting and I know that [bank] has strong high-yield and distressed debt desks."



Those would certainly be areas of particular interest, but again I know that many people come into the S&T with an idea of where they want to be and end up realizing another area is an even better fit so I'm open to exploration."

Do you think you'd rather be in sales or trading? Why?

For trading:

"I think I have a bit more of a quantitative skillset and while I am reasonably extroverted, perhaps not to the extent that I'd be that good at talking to clients all day and going out with them.

Ultimately, I think being in the weeds, seeing the flow in the market, helping to manage a book down the road, and all of that is something I'd really enjoy doing. I also think my extroversion could help in communicating with quants and sales people who may be affiliated with the desk; trying to bridge the gap between what traders need and what quants and sales people can help provide. So, I think trading would likely be the right fit."

For sales:

"I'm a very extroverted person who really enjoys talking to smart, interesting people. Obviously being in sales is not all roses; clients will get mad from time to time! But I think I would really enjoy a role that has me interfacing with clients, traders, and even members of the quant team should that be relevant."

Have you talked to many people who work in the industry?

Part of the reason why networking is so important is that you'll almost certainly be asked if you've talked to people in the industry and thus have a modest feel for what S&T is in practice.

"Absolutely. I've talked to quite a few people in S&T across a number of different firms, including some people here at [bank].

Ultimately, I tried to cast a reasonably wide net and talk to people across desks and across roles in order to try to best understand what S&T is practically speaking and how my skillset and personality may mesh with different desks."

What do you think makes a great trader or sales person?

For sales:

"I think a great sales person probably needs to be able to manage people just like a trader manages risk. By this I mean understanding who the clients are, what they



like to do, and ensuring when things don't go their way (either via a bad trade, bad pricing of a potential trade, etc.) that they don't have animosity towards the firm.

A great sales person also probably has a keen understanding of the market and broad opinions that clients would be interested in hearing about."

For trading:

"I think a great trader would be someone who is an excellent risk manager above all else. This probably means not only that they have the kind of exposure they want in their book, but also that they understand their risk fully, which is probably hard to do in some asset classes when you have so much going on.

A great trader probably also has to have the capacity to deal with ups and downs and be relentlessly focused for long stretches of time."

What do you think the role of a summer analyst is in S&T?

"I think the role of a summer analyst is to learn, be curious, find your fit, and not get in the way. Ultimately, I think people need to see your personality and you need to study where you fit best within S&T while at the same time understanding your value add is not going to be overly large so you shouldn't be getting in the way."

What do you think are the best attributes for a summer analyst in S&T?

"Being curious and asking smart questions, relentlessly trying to find where you fit best within the firm and what role is right for you, and to have the emotional intelligence to know when you should be asking questions and when it's best to just observe. Of course, it's an added bonus if you can do something practically to help the desk, but I understand not all desks will necessarily have work that a summer analyst can help on."

Why do you want an offer specifically from this bank?

Ultimately, you should spend some time – as covered in the Overview guide – seeing what the bank is best known for.

With that being said, giving a standard boiler plate answer is more than fine. It's really how articulate you are that matters for this question, not exactly what you say. Everyone knows that what you really want is an offer anywhere; you don't care *that* much about from what bank.

"I think it's a combination of a few things. First of all, looking at league tables I know this bank is strong across a number of different products and I suppose that's always a good thing.



Second, and I think more importantly, I've found everyone that I've talked to here really enjoyed their summer and full-time experience and I think at the end of the day that counts the most. You don't want to do something that will make you miserable just because it's prestigious or sounds good to your friends. So, culture and job satisfaction do matter to me and folks here seem to have both."

Tell me about how you keep up to date with the markets?

Here you need to be careful not to go overboard. If you say, for example, that your school has a Bloomberg terminal and you frequently go on it then you will get follow up questions about what functions you use on the terminal and how you access them. If you don't know these, then you will look like you're probably exaggerating how closely you follow markets.

It's a good idea to be as specific as possible, but make sure you can answer follow up questions about what exactly you've read and what you've taken from it!

"I try to keep up with markets as best I can while balancing college. I always try to skim the markets section of the Wall Street Journal and the Financial Times and read stories that pull my interest.

I then also try to find research reports from banks on just about anything and enjoy hearing a more opinionated view about not only what has happened, but will happen. I also read a few newsletters everyday, like from Matt Levine of Bloomberg."

It's also a good idea to talk about research reports you've read coming out of investment banks. You can find some of these by just Googling things like, "Goldman Sachs [equities/rates/FX] research report".

Just make sure, once again, that if you bring up a certain researcher at a bank that you follow that you're prepared to talk a bit about what they've said.

Note: Matt Levin's daily newsletter is very good and you should sign up (it's entirely free, you don't need a Bloomberg subscription or access to the terminal).

Can you give me an example of some story you've read recently and why you found it interesting?

You should have one or two recent news articles you've read; from Bloomberg, WSJ, or the FT that you can speak to.

Generally, it's a good idea to find one about global macro. For example, around what's happening in rates.

"I read an interesting piece from the WSJ about how a Biden election was boosting the long-end of the yield curve in anticipation that we're going to have less fiscal stimulus and more Fed actions under a Biden administration, which will mean the



low end will be stuck at the zero lower bound (despite inflation maybe picking up leading to a steepening of the yield curve).

Goldman Sachs also came out with their 2021 expectations report, which also signaled the likelihood of a steeper yield curve in 2021. Of course, many people have been anticipating steeper yield curves for years now and for the most part it has just got flatter!

I do wonder if the Fed will start engaging in more targeted yield curve control to try to suppress down any steepening if it does begin to happen. That would seem like a logical next step given how their conventional policy tools have largely been taken as far as they can go."

This is a good answer in that it shows you understand what you read and you developed a little side opinion (it doesn't matter if it's right!) on your own.

Let's say you're on the desk and you observe something that you think is unethical or wrong (either that someone did or said). What would you do?

Ethical questions are very common at larger banks. At Goldman you are almost always going to get one of these and they're trickier to answer well than you may think.

"Well, first of all I would want to make sure that I understand the internal compliance procedures of [bank] and if there is a proper channel you're supposed to work through.

Speaking more generally, I think what you would want to do is approach your boss – maybe a VP or MD – who you have some level of relationship with. Then you want to describe the scenario, but not the specific person, to gauge whether or not what occurred is normal or something that is actually unethical or wrong. Then, if it is unethical or wrong, they will understand how to handle the situation and what the appropriate next steps are."

If someone on the desk did something inappropriate, not necessarily illegal, how would you seek to deal with it?

"As a very young person on the desk, I think it would be inappropriate for you to get solely involved in trying to remedy the action and instead seek to inform someone better positioned to handle the situation.

I think it's always important to understand you don't have perfect contextual knowledge and where possible seek to make those in a position to handle the situation aware of it."



Let's say that someone you know is sharing all the answers for an upcoming exam with other students in a class because they somehow got them early (or they're the same as past years). What would you do?

"I think you need to take a step back and think about the fundamental fairness of this. Obviously, if these answers were shared with everyone then the professor would change the exam (lest everyone get 100% on it), so if these answers weren't shared with everyone then it gives an unethical, unfair advantage to those who do have these answers.

I would let the professor know about these actions and allow him or her to make a determination as to the next steps as it is ultimately the professor who has the authority to determine what can and should be done."

Let's say a client calls you and starts to scream about something. How would you handle that?

"I think part of any job is that you will come into contact with those who are angry about a situation (perhaps not of your own doing) and you will bear the brunt of it.

Like with any conflict, you want to seek to understand their perspective, offer your perspective, and generally let them vent it off. As a representative of the firm dealing with a client you don't want to end up in a situation where a client never works with [bank] again because of your actions or because you lose your cool."

What makes for effective communication in your view?

"Effective communication comes down to understanding the level of the person you're talking to and catering your language to their level while being clear, conscious, and never demeaning.

For example, if I asked you to describe your job you could get very technical, very quickly and it would all be way over my head. Effective communication would involve you understanding I don't know nearly as much as you do, so you need to tamp down the terminology for me."

If I were to call your prior boss, what would he or she say about you?

"Well, I think they would say I'm a very hard worker, easy to get along with, and go out of my way to try to understand my place and my role in the firm at any given time.

What I mean by that latter part is that I try not to get in the way when things are hectic and I can't be of help, but try to be of help and involved when possible."



What's your greatest strength?

You should pick what is *really* your greatest strength. Below are a list of ones that I think always go over well.

- Work ethic
- Determination to do the job even if it's ambiguous or difficult
- Considerate of what one's role is and how to succeed within it
- Ability to get along with a diverse group of people
- Not easily overwhelmed, relish and enjoy a challenge

What's your greatest weakness?

This is always difficult to answer because you don't want to say anything that would preclude you from getting the job (e.g. by saying you're very lazy).

Here are a few:

- You're very detailed oriented and sometimes spend too much time working on what should be short, simple tasks to make sure it's 100% correct
- Sometimes you get lost in the weeds of a task and forget the bigger picture
- You're very extroverted and sometimes forget that other people are more introverted
 - This is good to say if you want a sales role
- You're more introverted and definitely don't have an outgoing "sales" personality
 - This is good to say if you want a trader or quant-style role

Tell me about an accomplishment of yours that isn't on your resume?

You should always have one or two accomplishments to talk about that are not line items on your resume.

This could be an athletic thing (run a half marathon, even though you hate running), an academic thing (got an A on a paper even though you hated writing it), or a professional thing (completed a difficult task that's not on your resume).

Tell me about a time you worked in a collaborative environment?

You should likewise be able to talk about when you were involved in a team setting and preferably led the team. This will probably be an academic or athletic example. Make sure that it has occurred in the past few years (remember: nothing from High School unless it's exceptional).



Tell me about a failure of yours? What did you learn from it?

You should be ready to talk about a failure. A good one involves not meeting a deadline (either self-imposed or from your job / college) and that you learned the importance of budgeting your time.

"In my last internship I was given a task that I thought would only take a few hours but ended up being quite a bit more involved than I thought."

I missed the deadline. It wasn't the end of the world, but it was embarrassing for me. I learned of the need to budget my time appropriately, think hard about how long things will actually take, and be open and honest with more senior people when I'm struggling to meet a deadline so they know it will be delayed."

What character trait have you been trying to improve recently, if any?

You should think about this question seriously and come up with a true answer. Below is an example:

"I'm a bit more introverted by nature and have been trying to up my extroversion. I've done this by getting involved with more clubs at [collegel] and trying to meet a wider set of people than I'm used to."

Where do you see yourself in five years?

"Well, hopefully here full-time! More seriously, I would love to be involved in the markets in some way, still learning, and hopefully feeling like I'm developing a deep knowledge in a certain segment of the market."

Why did you choose your college?

Once again, you can come up with your own answer based on your own experience. Even just joking that it was the best college to accept you is a perfectly reasonable answer!

"I think [college] blends academic-life and social-life very well. I think it's important that you attend a college that puts you in a position to be able to get great jobs – like this one – while at the same time not being holed up in a library during your earliest adult years."

What's your favorite course been?

You can pick any course you like as long as you have a thoughtful answer. In fact, I'd recommend picking one that's not finance oriented. Saying your favorite course was finance 101, economics 101, or accounting 101 is far too uninteresting to say.

My answer, for what it's worth, was essentially:



"I took a course on economic history that I really enjoyed. I think a problem with how economics is taught today is how "scientific" it tries to be.

This economic history course was really the only course in all of economics I took where we took a step back and asked, 'What is economics anyway?' We started with Aristotle and moved our way up to Adam Smith and all of the grading was around essays you wrote about your own thinking and interpretation of these old texts, instead of just memorizing formulas."

If you could make your college better in one way, what would that be?

Again, this is a more personal question. Easy answers would involve having more sport programs, more "school spirit", more interest in clubs, etc.

Your answers shouldn't be around how people should work harder or there should be more on campus recruiting or anything like that. These questions are meant to gauge that you're a well-rounded person not just some finance cog.

Tell me three people that you admire? Tell me three people living or dead you'd like to meet?

You should pick three historical people who you admire and have a quick rationale for why. It's not a great idea to say family members, or people alive today who may have politically divisive views, as that can get awkward.

My answers would be:

- John von Neumann
- Justice Robert Jackson
- Winston Churchill

What's your favorite movie?

This may seem like a trivial question and a waste of time to ask. Whenever this kind of question is asked, it's because they want to see if you give a terrible finance answer.

For example, if they ask what your favorite movie is and you say Wolf of Wall Street you are going to leave a bad impression (because that's not how the real-world of high finance operates and you likely have a cliched view of finance).

To combat you want to show that you're an interesting person. Pick a movie that has won some awards and is entirely non-finance.

Some of my favorites:



- Amadeus
- Das Boot
- Good Will Hunting

What's your favorite book?

As above, giving a bunch of finance books could be a bad idea. Saying Liars Poker is just a bit too cliched.

You should say books that you actually have read, enjoyed and that are reasonably interesting and serious. Do not pick an academic book, however.

Some of my favorites:

- Anti-Intellectualism in American Life
- The Metaphysical Club
- George F. Kennan: An American Life

Tell me about a few books you've read recently?

For this question, you can say a few finance books if that's what you've read most recently.

If you're in the search for good business / finance books, be sure to look at the annual Financial Times / McKinsey top book awards.

If you could do anything else besides S&T, what would you do?

You should always give an answer that's in the finance industry here. Saying if S&T didn't work out then you'd go work at a non-profit somewhere raises a few red flags!

"Well, I'd want to stay connected to the markets in some way; hopefully at a bank like [bank you're interviewing at]. The obvious answer would be either investment banking or research.

I think research would probably be best for me as it's a bit more tethered to the market. So, I'd probably look to either get into equity research or be part of one of the other research teams focusing on global macro."

Let's forget about finance. If you could do anything else outside of S&T, what would you do?

This is a frequent follow-up question. If you have more of a quant background (you can code), then just say software engineer. If you have more of a qualitative background, then saying you'd be a lawyer would be a good choice.



Describe the type of people you don't get along well with?

"Well, I don't consider myself to be that difficult to get along with or that bothered by people's personalities generally. If I had to pick one group of people, it would probably be those who have very decided opinions despite not knowing much about what they're talking about. They're usually pretty annoying to deal with."

How likely would you be to take an offer if given one? Do you think you'd come back full-time?

You should always say you'd accept an offer if given. Don't try to get cute about it.

"Absolutely I'd accept an offer. I've done my homework and wouldn't be interviewing here if I wasn't confident that I'd do well as a summer analyst, get a full-time offer, and come back."

Where else are you interviewing?

You may get asked this question at the end. It's almost always because they're interested in giving you an offer.

It's an awkward question, but you should answer it honestly *a/ways*. It's a small world and you don't want your interviewer calling up friends asking about your status anywhere.

If you aren't interviewing elsewhere, trust me that it won't be a ding against you.

Market-Based Questions

Market-based questions are questions that deal with the market as it is today and are meant to figure out whether or not you really have an interest in markets.

Obviously, markets are incredibly broad and as such no one expects you to be an expert in all areas or to follow all areas closely. Instead the expectation is that you can talk at least about some area of the markets a bit.

It's important to note that my answers are meant to be templates for your own answers, because obviously markets change over time! Don't say the ten-year rate is eighty basis points (0.80%) if it's really 200 (2.00%) when you're reading this!

With that being said, these template answers will work regardless of where things are specifically trading now as they show you the approach to answering these questions.



What is one major asset class you're following closely?

"I'm quite interested in where rates are right now. In particular, we've seen a steepening out of the yield curve, which I think is indicative of the fact that market anticipates some level of enhanced inflation (5y5y has ticked up as well) and stronger economic growth moving forward while the Fed holds the overnight rate at the zero lower bound."

Are markets really that efficient?

"I certainly believe that if the conditions of the Efficient Market Hypothesis hold true then markets could be efficient. However, I think that it's hard to imagine many markets – unless they are very liquid, very easily accessible, and very transparent – could ever really be considered efficient."

What's the mandate of the Federal Reserve? Are they living up to it?

"The Fed has a dual mandate of price stability – considered to be keeping inflation below 2% - and maximum employment. Certainly, the Fed has done a good job on the first front over the past number of decades. On the latter front, I would argue that the Fed alone can't generate maximum employment without fiscal policy being aligned. Certainly in 2020/2021 there's room to go on getting to maximum employment (which is probably around a 3% unemployment rate)."

What are some of the major central banks and who are their chairs / governors?

- The Federal Reserve – Jerome Powell
- The European Central Bank – Christine Lagarde
- The Bank of England – Andrew Bailey
- The Bank of Japan – Haruhiko Kuroda
- The Bank of Canada – Tiff Macklem

What routine news events have a large impact on rates?

"Federal Open Market Committee meetings, Speeches from Federal Reserve board of governors / the Fed Chair, Non-Farm Payrolls, Month / Year End dates due to rebalancing, and global shocks (wars, economic crashes, etc.) all can move rates."

What's the Fed Funds rate right now? Do you think it'll stay there for a long time?

"The target range is 0-0.25% (or 0 to 25 basis points). Fed Funds futures aren't pricing in a rate hike for the next year and the two year is at roughly 14 basis points so it's certainly unlikely we'll see Fed movement from these levels for a few years."

<https://www.cmegroup.com/trading/interest-rates/stir/30-day-federal-fund.html>



The reason rate hikes could occur, of course, is if we end up getting significant inflation – perhaps due to the increase in the money supply as a result of COVID stimulus – or we get down to full employment quickly.”

How does the monetary policy of the U.S. differ from other major economies right now?

“In response to COVID, the major central banks began to largely converge on the same set of policy tools being utilized. This means getting the overnight rate down to roughly zero and purchasing large quantities of government issued bonds.

One element of the Fed response that was not replicated by all others was the purchase of high-yield bond ETFs. This was done to try to create a favorable issuance environment for corporate issuers in the midst of the turmoil by essentially backstopping corporate bonds (ensuring there was a buyer there, so their prices didn't dip too low). Indeed, the Fed even explicitly said they would participate in some new issuances by corporates if the terms were amenable. This led to a great increase in confidence for market participants who routinely deal in the corporate bond market.”

<https://www.etfstrategy.com/federal-reserve-expands-asset-purchases-to-high-yield-etfs-94578/>

How do you follow what rates are doing?

“I find following rates to be a bit more difficult than equities. I suppose because it's an asset class that less people on Mainstreet are routinely trading. I primarily look at the rates area of the WSJ and try to keep up on global macro via Bloomberg.

I also have really enjoyed the writing of Zoltan Pozsar of Credit Suisse. He gets into the minutia of money markets in particular and while I can't say I understand it all, I've really enjoyed trying to learn what he's talking about in his research reports.”

When we talk about the stock market, what are we really referring to?

“The stock market is obviously a broad term that encompasses a number of exchanges both domestically and around the world. I suppose when most people talk about the stock market they are talking about S&P 500, the Dow Jones, or the Nasdaq domestically. Internationally, they may be referring to the FTSE (London Stock Exchange) or Nikkei (Tokyo stock exchange).”

What do you think is driving the stock market right now?

If there are specific things happening in the markets over the past week that are clearly driving them, you can talk about them. However, it's always good to answer this question



by showing that you understand what partly drives equity markets in a bit more fundamental of a sense.

"Well, I think in a general sense when you think about equity valuations you can just revert back to thinking about discounted cash flows. What we've seen over the past four years is a decreased corporate tax rate (positive for free cash flow) and now lowering rates (so a lower discount rate on cash flows generally). These combine to boost cash flows overtime and spur valuations higher.

Now, of course, many of the technology growth stocks aren't exactly throwing off free cash flow (never mind profits) so there's certainly a growth story underlying recent market moves over the past few years in some sectors.

I think you can also make an argument that with all of fixed income yielding so little, equities are an almost de facto risk-adjusted safe haven because at least there's a possibility of some return in excess of inflation. This is borne out by the decoupling of bonds as a natural hedge over the past few years for equities, which I think raises broader concerns about portfolio positioning of pension funds and endowments that are trying to keep a reasonably steady return rate."

Where is oil at right now? What's driving it?

"Well, WTI spot is at about \$40 although I'm not sure where some of the other indices are at exactly like Brent Crude; although I know it's normally quite a bit higher.

Right now, from what I've read although I don't follow oil that closely, it appears as though the U.S. rig count is beginning to pick up at perhaps a faster pace than demand is accelerating. Of course, international oil supply also matters, but I haven't heard about material changes to supply but that could also be a factor for prices staying suppressed and the recent declines we've had off of already low levels."

Rig count: https://ycharts.com/indicators/us_rotary_rigs

WTI: https://ycharts.com/indicators/wti_crude_oil_spot_price

What's the 10-year at and where do you think it will go?

"The 10-year is at roughly 78 basis points (0.78%) currently. Yields have ticked up over the past few months – around 20 basis points - inline with the general steepening of the curve in that area. If we look at 1Y10Y forward rates – what you can lock in for a 10-year yield one year from now – we see it up roughly 10 basis points (0.10%) so the market is pricing in a bit of a continued rally in yields.

This strikes me as probably being correct due to underlying inflation fears I have due to the increase in the money supply. One thing that could offset this is that if we see a strong rally over the next six months in 10-year yields, perhaps the Fed would start



to engage in active yield curve control (which, to date, they haven't yet) trying to get the curve flatter across the 2s10s and 10s30s."

10-year yield: <https://fred.stlouisfed.org/series/DGS10>

What's the yield curve look like? Thoughts on it?

"The yield curve is an interpolated curve showing the relationship between yield and maturity. The curve is interpolated because in reality there are not bonds at every single possible maturity point (e.g. maturities of exactly 892 days away, for example).

Right now the yield curve is quite steep and obviously upward slopping. I think that we could end up seeing the Fed intervene to try to get the curve lower and flatter as yet another way of trying to spur economic growth."

What's the reason why the Fed (or anyone else) cares about having 30-year Treasury bonds lower?

"The point is that much of what occurs on "Mainstreet" is benchmarked off of Treasuries. Having lower Treasury yields reduces the cost of funding indirectly of corporate credit, which in turn spurs economic growth. Some also suggest that making Treasuries less attractive to hold – due to lower coupons – pushes folks into lending to "riskier" counterparties like corporates, which spurs economic growth as well."

What do you think about credit markets right now?

"Well, the BofA AAA effective yield index is at 1.60%, which is down around 90 basis points from this time last year. Obviously, we had a spike around COVID, but the Fed actions around buying corporate bonds – including new issuances of corporate bonds – of an investment grade level seems to have not only calmed markets, but boosted issuance and suppressed yields.

Indeed, if we go all the way out to BBB – which is the demarcation between IG and high yield – we still are looking at 2.34%, which is astoundingly low and down around 100 basis points from this point last year.

High yield indexes from BofA are around 5% and for CCC+ and lower – where you have the highest default probability – they are around 11%, both of which are also down YoY.

These levels across credit are all very tight - although not as tight as during their tightest in the summer of 2020 - and signal strong credit markets and lots of cash on the sidelines ready to be put to work even in risky credits.



I think we'll see a further normalization of these yields as stimulus fades and as it's realized that the Fed can't (and won't want to) backstop individual high yield names, but instead perhaps just keep buying index funds to suppress general yields and make it a more favorable issuance environment for credits that can go out and raise new funds."

High yield effective yield: <https://fred.stlouisfed.org/series/BAMLH0A0HYM2EY>

BBB effective yield: https://ycharts.com/indicators/us_corporate_bbb_effective_yield

AAA effective yield: https://ycharts.com/indicators/us_corporate_aaa_effective_yield

Understanding what effective yield is:

<https://www.investopedia.com/terms/e/effectivelyield.asp>

What kind of credit products are out there? What one interests you most?

"Credit is obviously a very diverse term that can encompass corporate debt throughout the capital structure from investment grade bonds to distressed bonds, loans, CLOs, and credit default swaps.

I personally find the high yield and distressed debt space quite interesting because you have a mix of fast money and institutional buyers, more illiquidity than in IG, and obviously quite a lot of volatility. I think it's probably an area where you're going to see a greater diversity of outcomes given the fact that some names you hold – especially if they're CCC+ - are going to end up in some kind of restructuring process (including in-court bankruptcy).

Obviously, I don't know what I don't know, however. So I'd be open to exploring any area of credit if I were lucky enough to have the opportunity."

How do you stay up to date on credit?

"I find staying up to date with credit – like rates – to be a bit more difficult than equity. I suppose this is because your average person won't be able to go buy credit default swaps or even distressed debt.

I try to stay up to date by following investment grade and high yield indexes and seeing how they converge or diverge, reading WSJ and Bloomberg articles about credit when they pop up, and trying to think about how credit is moving given economic news that comes out.

I don't really follow any specific credit from any specific company, but just try to observe the market more broadly."



What are the major FX rates? Any major trends in FX?

"The major crosses are EUR/USD = 1.19, USD/JPY = 104, and GBP/USD = 1.31.

One major trend in FX is hedge funds betting against the dollar (dollar weakness) in the futures market, which has reached the highest level in over a decade. The reason for the bets on dollar weakness surround a relatively weaker economic recovery for the US than in Europe and the anticipation that interest rates will be at zero for a prolonged period of time."

You can find major FX trends by just doing a quick Google search and summarizing what traders are doing most in the market: <https://www.reuters.com/article/usa-markets-dollar-analysis-idUSKBN25S3KN>

Why do currencies fluctuate?

"Obviously in FX everything is relative; a currency is only "strong" or "weak" in relation to the currencies it trades against. So, what drives strength or weakness will be relative differences in monetary policy (interest rates and central banking policy), economic statistics (inflation, economic growth, etc.), politics (political stability, foreign policy stances, domestic spending objectives, etc.)."

How do you stay up to date on FX?

"Generally, I don't follow FX quite as much as I do other asset categories, but I find Reuters and the Financial Times have good coverage of FX and I try to stay up to date on general trends.

I think FX is probably one of those assets where you'd learn more being on a desk for a week – seeing the flow of the market – than you would spending months reading articles on FX."

What's one overall trend in markets you're following closely?

Take a spin through the WSJ home page and you'll find many market trends. These trends should not be too esoteric as a likely follow-up question would be how you would take advantage of this trend! Keep your answers simple, informed and under two minutes.

"An interesting trend that appears to be beginning is the rally in financial stocks, which have been hammered in 2020 and are still down 17% (according to the largest financial stock index) to date.

The rally appears to come off of the back of lower than expected losses on their loan portfolios and strong ECM/DCM activity for the large banks. For the financial stocks that cater towards retail investors – like Schaub – there has been a large influx of new customers and deposits.



A primary drive for a continued rally would be that net interest income – the base layer of profitability for banks – can likely only increase from these levels and traditional investment banking (M&A fees) should pick up moving forward after deals were largely put on pause.

The old saying is that you should never bet on financials or energy, I know, but it does seem like financials are one area of the market that still has not recovered (but is in far better shape than what folks thought they would be in in March / April)."

If you were to take advantage of this market trend, how would you do it?

"The easiest way would be to look at indexes that focus on the financial sector and then ensure that the stocks that comprise them are aligned with your underlying thesis. So, for example, XLF is comprised of most of the large bank with ECM/DCM/IB/Trading along with a few asset managers.

You could drill down further on specific names and invest individually, but I haven't dug too deep into this trend. It's more something I've observed that I think is quite interesting."

What are some important economic indicators to watch out for?

"In terms of things you should probably keep your eyes on, even if it's not viewed as being directly relevant to the product you're involved with, I'd say they would be: GDP, CPI, moves in 5Y5Y swap rates, initial jobless claims, non-farm payrolls, PMI, moves in VIX, high yield and IG index trends, moves in the 10-year yield, and the S&P 500 level."

If you could only follow one economic indicator what would it be?

"This probably depends on the product you're trading; I think generally inflation indicators will have a large affect throughout every asset class because changes in inflation – especially sharp changes – will change rates and rates in turn will change credit and equity valuations.

So, if you had inflation tick above 2%, you'd then expect the Fed to raise interest rates, rates to rise (in yield terms) across the curve, the curve to likely flatten, equity valuations to potentially decline, and credit to trade down (meaning yields increase)."

How do you think about equity valuations right now?

For this question, you should think in terms of current P/E ratios of indices and forward P/E ratios of indices and then offer a short rationale for the discrepancy between the two.



"Well, in order to determine this, I think I would look at the current P/E ratios of major indices and then the forward P/E ratios as well. Currently, it's true that we're at the highest P/E ratios since the great financial crisis. However, that's a bit of a mirage because earnings have been suppressed as the result of COVID and just because earnings in one year go down doesn't mean a stock should go down by 20% (given that a stock should be valued more or less on the present value of all future cash flows).

The current S&P 500 P/E ratio is 36, however the forward P/E ratio is 24, which gives credence to what I mentioned before about the current P/E ratio being a bit of a mirage.

P/E ratios of between 18-24 have been observed over the past six years, so 24 may be a bit high, but it's unlikely as even the forward P/E ratio is likely accounting for lower earnings that would otherwise occur without COVID so I don't think equity valuations are necessarily stretched or outrageous right now."

What's generally the relationship between bonds and equities?

"Historically a well-balanced portfolio would be comprised of both equities and bonds. When equities sell off then bonds would often rally (in price terms) with their yields falling.

This created a kind of offset that reduced down the volatility of one's portfolio. However, over the past year this correlation has begun to turn from negative to positive; meaning when you have a selloff in equities often that comes along with a selloff in bonds (in price terms) too.

So, it doesn't strike me as there being a hard and fast relationship anymore. Of course, the correlation could turn negative once again, however!"

<https://www.bloomberquint.com/gadfly/government-bonds-may-not-save-your-portfolio-buy-volatility>

What's the relationship between bond prices and yield?

As bond prices decline, yields increase.

If we have a bond with a price of 90, 10% coupon, and it matures next year, then what is the yield to maturity (YTM)? What about if maturity is two years from now? What's the current yield?

To get the YTM for a bond maturing in just one year, then you can use the formula: $YTM = (\text{coupons} + (\text{face value} - \text{current price})) / \text{current price}$, which gives you: $20/90$ or 22.22% (you don't have to do the mental math to get the exact percent, as long as you get the formula right).



For calculating maturities longer than one year, you need to use the estimated YTM formula (which doesn't get you an exact number, but close to it).



Yield to Maturity =
$$\frac{C + \frac{F - P}{n}}{\frac{F + P}{2}}$$

Using this formula – you can work it out on paper, you don't need to do it in your head! – you'll get 15.789%.

The current yield is just 10 (the amount of coupon income you get per year) divided by 90 (the current price of the bond), which would be 10/90 or 11.11%.

How do I hedge my exposure in rates trading? Equities derivatives trading?

"In rates trading this is primary done by getting your DV01 of a given position down to zero by either buying or selling offsetting Treasury futures positions.

For equity trading this is done via delta hedging, which involves going long or short the right amount of the underlying shares in order to be perfectly market neutral."

What are various measures of return?

"While there are many, some of the most popular are the internal rate of return (IRR), multiple on invested capital (MOIC), and the Sharpe ratio."

Where do you see gold going? Are there any more commodities that you follow closely?

It's generally a good idea to know the price of gold and have some simple thesis about it but knowing about any other commodities (outside of oil) is entirely unnecessary. No one expects you to have a deep knowledge of agricultural futures unless you grew up on a farm in the Midwest!

"The price of gold is currently hovering around \$2,000, which is a roughly 30% increase YoY. Interestingly, gold like almost every other asset class fell in the wake of COVID before rallying stronger than most other asset classes (outside of some tech stocks).

Traditionally gold has been viewed as some form of inflationary hedge – although, of course, it hasn't always moved that way in any given week - and the rally is likely due to investors thinking that equity values are stretched, and the dollar will continue



to weaken. One interesting thing I've observed is that while bets against the dollar (on the dollar devaluing) have increased, gold has increased in value significantly.

I'd say this may be a trade you'd look at fading as it does seem rather consensus and I'm not entirely sure we're going to see the U.S. underperform competing economies over the next few years. So, we may have dollar strength and weakening gold prices moving forward.

I don't really follow any other commodities outside of loosely following gold and oil; I can't claim to be an expert on soybeans, for example!"

What are some alternative asset classes? How are they doing right now?

"Alternative asset classes include things like investing in private equity, hedge funds, real estate, commodities, Bitcoin or crypto currencies, art, collectibles, etc. I don't know that there's a strict definition of alternative asset classes beyond them not being traditional asset classes and generally being more opaque with more volatile returns potentially.

With that said, obviously many institutional investors now have significant dollars invested in private equity or hedge funds strategies; some are even buying Bitcoin!

Ultimately, alternative asset classes seem to be correlated quite strongly with traditional asset classes over the past few years. For example, in the wake of COVID you may have anticipated that Bitcoin – a decentralized currency with a limited supply – would act as a bit of a safe haven, but it fell like everything else before rallying along with everything else!"

Investment-Based Questions

Investment-based questions are those that rely on you having to pitch an investment. There are only a few investment-based questions that you will possibly be asked during an interview.

These questions only come up in superdays. It's also important to keep in mind that your interview is just 20-25 minutes. No one is expecting you (or wants you) to launch into a five-minute pitch as that would take up an exorbitant amount of the interview.

What they're looking for is that you a) are prepared b) know some of the critical elements of answering these questions and c) speak clearly and coherently.

Your ideas also don't have to be original. If you read of an investment idea that you think is quite good (on SeekingAlpha or someplace else) feel free to think about it, put it in your own words, and use it.



If you had \$1,000,000 personally how would you invest it?

"Well, I suppose the conventional answer would be in a diversified portfolio of stocks and bonds; probably 60/40. However, what we've seen over the past year is not only bond yields at record low levels, but also them being correlated positively with stocks mitigating the kind of risk off-setting underlying that kind of portfolio construction.

Because of how young I am, I'm not overly worried about a 40% correction in my portfolio at any given time (since I'm not relying on drawing out a certain percent per year). So instead of being quite heavy on bonds, I would probably be 50% S&P 500, 20% emerging market stocks, 10% investment grade bonds, 10% Treasuries, and 10% gold.

In this kind of portfolio gold would operate (hopefully!) as a replacement of holding assets denominated in non-USD and as a general slight hedge against market underperformance."

If someone gave you \$1,000,000 and wanted a constant stream of income every year for the foreseeable future, how would you invest it? What would be a reasonable amount per year for them to reliability get?

"For this amount of money, it would probably make sense to look to a portfolio of municipal bonds, corporate bonds, Treasuries, and S&P 500 index funds. Perhaps your mix would be 50%, 20%, 10%, 20%, respectively.

If the individual has a current income – perhaps they are still employed – then Muni bonds can help by reducing your overall tax liability. Further, this portfolio of Munis, corporate bonds, and Treasuries can get a reliable amount of coupon income coming in per year (I'm assuming the S&P 500 dividends are reinvested back into the fund, as most ETFs do automatically). The S&P 500 helps with the overall growth in assets of the money invested and adds some greater volatility to the mix."

Pitch me a stock?

Getting asked to pitch a stock is the third most common investment-based question to be asked. This is surprising to some people, but that's because they fundamentally misunderstand what S&T is.

Remember, the vast majority of desks have *nothing* to do with equity trading whatsoever. Further, most of the people who actually work in the equity space are not dealing with cash equities, they're dealing with equity derivatives (index or single name). Finally, no one is sitting around the trading floor picking stocks; any equity flow is fundamentally driven by client activity.



The real reason why this question is ever asked is just because it used to be asked quite a bit ten years ago and some think it's a good way to gauge your presentation skills. The reality is this question shouldn't even be asked as picking and pitching stocks long/short is irrelevant to almost any job on the trading floor (obviously it's applicable to equity research, though, but that's distinct from S&T!).

With all that being said, it's still worthwhile preparing a stock to pitch. You don't need to model out a DCF or come up with some esoteric play. Here's what you need to do:

- Choose a name that's familiar, but not trendy (don't pick Tesla!)
- Pick a company that actually has reasonable traditional financial metrics (not some growth tech stock)
- Make sure to look over their most recent earnings call deck (see link below for an example)
- Try to see if you can find any research reports from banks on the stock (if you have access to a Bloomberg terminal they'll be there, if not you'll be able to find articles referencing them online)
- Know some key financial metrics such as the EBIT or EBITDA margin, FCF, net profit, EPS, and P / E ratio
- Know where the stock is trading now, where it was six months ago, and where you think it'll go (profit target)
- Have an underlying thesis that's multi-faceted for why the stock will move up (in your view)
- Have a stop-loss (point at which you get out of the trade) and a rationale for why it may go down
- Mention what the leverage ratios are to show that you understand that equity is inextricably linked to credit conditions of the company as well
- Make sure to spend no more than two to three minutes going over your thesis
- Don't be the victim of just blurting out an endless stream of numbers, this question is more about your presentation skills; no one thinks you'll be able to give an utterly convincing and technical pitch in just a few minutes

"I think Ford is actually quite an interesting stock, because it's really both a financial play due to Ford Credit and an automotive play.

Ford released their earnings a few weeks ago with \$34.7bln in revenue – up 3% YoY – and EPS of \$0.60, both of which beat street estimates.

Ford pulled out \$6.3bln in FCF on a EBIT margin of nearly 10%, which was boosted by Ford Credit which handles the leasing and financing side of Ford vehicle sales.

Ford currently has \$45.5bln in liquidity – defined as available cash and revolver capacity – with low leverage and coverage ratios on its debt relative to its peers.



I think Ford is underpriced at present because of continued uncertainty about the capacity for consumers to continue to meet their lease and financing obligations, despite no indication of loan losses anywhere near what was expected.

I also think the market is not adequately pricing in the soft restructuring that Ford is still pursuing that will see it's mid-size sedans phased out, some plants likely closed, and a focus on what Ford is a large market-leader in: their SUVs and F-series trucks.

Finally, I think the market is discounting the new product offerings – including an all-electric SUV and the newly introduced Bronco – which are entirely unique offerings in the North American market that have already sold out their limited initial run.

So, I think you can think about Ford as really being:

- *A bet on the resiliency of the North American consumer (which will be expressed via Ford Credit strength)*
- *A bet on slimming Ford down to what they do best working out*
- *A bet on the introduction on new vehicles that demonstrated market demand via already selling out their first run*

Ford's stock is currently at \$7.50 and I think it has room to move up to north of \$10 within the next calendar year. Ultimately, outside of the pandemic, Ford has not been below where it's trading now since the great financial crisis. This is despite the strong liquidity profile, limited stress on its debt covenants, and strong earnings just reported. The stop-loss I would set at \$5, which Ford was last at during the spring of 2009.

It doesn't hurt that [bank] also increased their price target 30% on Ford."

Find earnings slides: <https://shareholder.ford.com/investors/financials-and-filings/default.aspx>

It's always a good idea at the end of your pitch to note that a certain bank's equity research folks have upgraded the stock or put a high price target on it (as that lends some credibility to your pitch).

Pitch me a stock to short?

Pitching a stock to short is no different than pitching a stock to go long in terms of what you should cover. There are a few things to add, however.

First, and I can't stress this enough, do *not* pitch shorting some hot tech stock like Tesla. That would give the impression that you are spending a lot of time on Robinhood or something.

Second, you should have a rough idea of the borrowing cost. Unlike when you go long a company, there are costs to going short since you are technically borrowing those shares.



For example, if the borrowing cost is 10% and you buy 100 shares of X at \$110, then you are paying $(0.10 \times \$11,000)/365 = \3.01 per day to maintain that short (ignoring the actual PnL on your position).

Part of the reason for not picking a hot stock is that the amount of short interest outstanding – like for Tesla – will be quite large so even if you're directionally correct in your bet, you may bleed away profits via the borrowing cost of your trade.

"Gamestop I think is a really interesting stock. Obviously, like most retailers, it has been in structural decline for the past five years with its stock falling from the \$40s in 2015 to around \$12 today.

From mid-2019 to mid-2020 the stock was rangebound between \$3 and \$7 but has now rallied significantly to \$12. There are a few reasons for this.

First, Gamestop undertook an out-of-court restructuring which saw nearly \$415m in Senior Notes coming due in 2021 – that Gamestop was unlikely to be able to refinance, due to its already precarious financial state – exchanged for new Senior Secured Notes due 2023. This pushed out the impetus for a Chapter 11 bankruptcy, in other words, and gave some breathing room for Gamestop to turn things around.

Second, in conjunction with the capital structure restructuring there has also been a restructuring of the Board and senior managers within the company. Most notably has been Ryan Cohen – founder of e-commerce company Chewy – buying a significant part of the company and pushing for the installation of new Board members focused on pushing Gamestop to a more digital-first strategy.

Third, the major consoles – Xbox and PlayStation – hadn't been updated significantly for the past five years, but have new versions coming out in late 2020. This is viewed as significantly revenue enhancing for Q4 2020 and Q1 2021.

Fundamentally, I think there's been significant irrational exuberance occurring with Gamestop. Gamestop still has \$500m in leases over the next five years that they can't discharge, as they could in Chapter 11, and now have nearly \$500m bonds coming due in 2023 with a blended average coupon rate of over 8.5%.

Right now, Gamestop is narrowly profitable with Q3 EPS of \$0.1-0.2, which it had forecasted to be \$1.15-1.30 prior to COVID. During this period, they saw net sales fall roughly 25%, which was also a larger fall than analysts expected.

What I draw from this is that a significant – but not astronomical – drop in revenue has led Gamestop to be only narrowly profitable. What does this portend for a future in which foot traffic to Gamestop locations – of which there are still over 500 – doesn't bounce back after COVID? Do forecasted sales for consoles dry up as holiday shoppers choose to buy online from retailers like Amazon or Best Buy?



These consoles also focus on digital games, downloaded online, not physical discs; what does that do for Gamestop moving forward?

I am bearish on all of these questions. While Gamestop had lots of short interest outstanding prior to the past month, because of its structural decline making it profitable to do so, many have been burned by the recent rally. Definitely looks like a short squeeze occurred. As a result, short interest has fallen and borrowing costs are more reasonable for a short term short trade on Gamestop.

I would set my profit loss at \$15 – which including borrowing costs would be more like \$16-16.5 over a two-month time horizon – and my profit taking level at \$5.5 per share, which is just shy of where it traded pre-COVID most often.”

Pitch me a global macro trade?

Getting asked for a global macro trade idea is certainly more relevant to S&T as coming up with trade ideas in FX or rates is a critical part of the job of a sales person and having conviction and positioning your book a certain way is one of the primary jobs of a trader.

It's quite rare you'll get asked for a global macro trade. Primarily because global macro trades are quite a bit more complex and esoteric than just going long or short a stock. However, it's a great idea to have a global macro trade idea with a reasonable level of specificity. That would be highly impressive.

Here are some general guidelines for your global macro pitch:

- Keep it short; 2-3 minutes at most
- Don't pick an overly technical concept like an options trade, because then the follow ups could quickly be very technical
- Don't come across as too self-assured; stay humble and say this is just an idea of yours
- Clearly state your entry and exit levels in the proper context (e.g. for rates the entry and exits would be in basis points)
- You should make sure to be explicit in terms of what the underlying economic or financial theory is that is driving your trade

“A major story line over the past few years has been when the yield curve has temporarily inverted, which happened for 2s10s in August of 2019. Prior to COVID the yield curve was still very flat as rates had inched up, but economic activity slowed. With COVID pummeling the front end of the curve, the curve obviously steepened out and we've seen 2s10s go from as low as twelve basis points to now around 70 basis points.

From the beginning of 2014 to when 2s10s inverted, we saw a consistent flattening of the yield curve and I think we're going to see the curve flatten from the modest steepness we have now.



I think this is primarily because like from 2014-2019 we're going to enter into a lower-for-longer phase with relatively anemic, but consistent economic growth. Further, with the low end of the yield curve as low as it can go – practically for the U.S. as negative interest rates have been something the Fed has said they will not consider – yield curve control could be the next phase of monetary policy. This would mean the Fed aiming to tamp down 2s10s and 10s30s.

I'd add greater credence to this given the likelihood of gridlock in Washington preventing large fiscal stimulus from continuing to flow, which adds more pressure on the Fed to do its part to achieve its dual mandate, which it could hope to do by have a low, flat yield curve.

I would look at initiating a 2s10s flattener – selling 2s and buying 10s – at 0.69 with a stop loss at 0.90 and a take profit level of 0.30, which is roughly where it was at the beginning of this year.”

2s10s chart: <https://fred.stlouisfed.org/series/T10Y2Y>

Some examples of rates trades (an older Credit Suisse report, but shows the kinds of trades that can be done): https://research-doc.credit-suisse.com/docView?language=ENG&format=PDF&source_id=csplusresearchcp&document_id=805810250&serialid=6YmPCV4P%2BqDpjH05ZtgV4fvFAGysdCUUXLyvk5SI9v0%3D

Here's a short research report from CS on them anticipating an inversion of 2s10s about a year prior to it happening: <https://plus.credit-suisse.com/rpc4/ravDocView?docid=V7dZub2AD-hTM>

You'll note that in a prior question in this guide I brought up the curve possibly steepening as an answer. Here I'm talking about it potentially flattening. What I'm hoping to get across here is that you don't need to give answers that are “right” when it comes to forecasts for the future, as no one knows what will really happen! Instead, you just need to give answers with substance that are interesting.

Interviewers don't care about the exact trades you'd do, or where you see markets going in the future, they care about how you speak about these things.

Questions for Interviewer

At the end of your first-round interview (if done via phone) or at the end of each one of your superday interviews, time will be reserved for you to ask your interviewer questions.

These questions are incredibly important. I'm continually shocked by the fact that many interviewees will ask no questions or ask questions about when they'll hear if they got the offer (don't ever ask that!).



You should only ask two or three questions to your interviewer at most. Mainly because there won't be much time left at the end of your interview, so you don't want to overstay your welcome.

Below are the five questions you should choose from or augment slightly.

What do you think makes for a great summer analyst? Is there anyone that really stands out in your mind?

This is always a great question to ask. Interviewers love giving their opinions on what makes for a great junior employee. I'd ask this question first in any interview.

There's obviously constant talk about how technology and automation are changing the floor. Obviously, it will change different desks to different degrees. But how do you see technology altering S&T moving forward?

This is a good question to ask someone who has a more quant-based role (such as an equity derivatives trader). Be careful not to ask this to someone who has likely had their desk negatively affected by automation (such as flow trading desks like cash equities).

Obviously post GFC there have been a lot of regulatory changes that have impacted the trading floor (most notably Dodd-Frank). Have all of these regulatory changes altered your view of S&T in any way moving forward?

This is a thoughtful question to ask that shows you understand how S&T has changed over the last decade.

Over the past five years it appears some banks have pulled back on the kind of risk they're willing to accept from a S&T division. At the same time during COVID you saw S&T divisions make up for the lower profits in traditional investment banking. Do you think we're going to have a swing back to S&T being given more risk-allowance?

This is another good question to ask that shows you understand that S&T divisions have not been given the same level of discretion as they have in the past.

As always, you don't want to sound too negative or pessimistic about S&T. However, showing that you aren't naïve and understand what you're getting into is invariably a good thing.



Obviously, you're probably going to be a bit biased. But generally, what do you think a young person should be looking for in desks in terms of attributes? More flow-based or more illiquid, larger desks or more niche, etc. Or is it purely based on one's personality and skillset?

This is another good question to ask. Interviewers love giving their opinions and asking them what they think makes for a great desk is a good way to show you're humble and open-minded about where you'll end up.

Product-Specific Questions

Below are some relatively basic product-specific questions that you could face in an interview.

These are all the most basic questions taken from the Desk Guides that you'll find in the members area.

In a first-round interview you will not get any of these questions. However, in a superday interview – especially if you express interest in a certain desk – you can expect some product-specific questions to crop up. It's a good idea to spend time with those Desk Guides if you have a certain area of the trading floor you'd really like to be on.

No one expects you to be an expert on a certain desk and the product traded there. But showing you are capable of answering these kinds of basic questions makes you *really* stand out and puts you in the best position possible to get an offer.

Note that the questions in the Desk Guides build off of each other. So, if you're a bit confused about any of these questions, check the relevant Desk Guide and the other questions and answers provided in the guide will likely clear things up.

Basic Credit Q&A

What are corporate bonds?

Bonds issued by corporations (as the name would imply!) that generally have maturities between 1-20 years (rarely sometimes longer) that have *fixed* coupons. Coupons are paid semi-annually on a 30/360 day-count convention.

What drives the pricing of corporate bonds?

Ultimately, as with everything in finance, it's the market clearing price (what a marginal individual is willing to pay). However, what makes those investing in corporate bonds feel comfortable receiving a certain coupon amount is a mix of:

- The credit rating given to the bond (from Moody's or S&P)



- The historical performance and credit worthiness of the corporation in question
- Where the bond resides in the capital structure
 - Is it secured? Unsecured? How much debt (bonds, loans, revolvers) are ahead of the bond in question
- What comparable bonds are trading at (or were issued at)

What's the credit spread?

The credit spread is the difference between the yield of a corporate bond and the yield of the underlying government bond (treasury, in the U.S.) of the same duration.

For example, T+250 on a 10-year corporate bond would mean it trades at the ten-year treasury note, plus 2.5% (250 basis points).

What makes credit spreads move?

Credit spreads will widen due to general economic downturns, the company performing poorly, or general pessimism in the credit markets. Credit spreads will tighten if the economy is healthy, the company is doing well, or credit markets are very active (lots of new issuance, new issuance pricing tight to treasuries, etc.)

What are Fallen Angles?

Fallen Angles are bonds that were issued as investment grade but have now fallen below investment grade. In 2020 Fallen Angles became a talking point given the rapid rise in them: <https://www.bloomberg.com/news/articles/2020-03-13/why-billions-in-bonds-now-trade-like-fallen-angels-quicktake>

What kinds of debt are most common in capital structures by seniority?

- Revolver (Revolving Credit Facility)
 - Like a credit card; can be drawn and paid down over the maturity period with some restrictions
- Term Loans
 - Can have Term Loan A or Term Loan B (TLBs are often syndicated and traded by a "bank loan desk" on the floor)
- Senior Notes
 - Either secured or unsecured
- Subordinated Notes
- Mezzanine Debt
- Preferred Equity
- Common Equity



What types of covenants will you likely find in corporate debt (in particular, corporate bonds)?

Debt docs are the documents at issuance that govern the terms of the debt being issued. These debt docs will contain covenants that specify the things that can't be done (negative covenants); if the company does break any of the covenants then the holders have a right to force the company (broadly speaking) into a technical default unless a resolution can be found.

- Coverage ratio (min EBITDA / interest expense)
- Leverage ratio (max debt / EBITDA)
- Restricted cash (min amount of cash that must be kept aside at all times)
- Negative pledge provisions (inability to pledge certain assets for new debt)
- Restrictions on guarantees that can be offered to subsidiaries without prior consent
- Payment priority in the event of asset sales
- Additional covenants to protect priority of the debt (meaning the place in the capital structure) and under what conditions additional debt can be placed ahead of this debt

Basic MBS Q&A

Philosophically, what really are MBS?

While there are many permutations, a MBS is a form of ownership of a wide swath of loans (mortgages) and the cashflows tied to them over time. The MBS owner is essentially the one being paid by the individual holding the mortgage (without the mortgage holder knowing it, of course). Every month the MBS holder will receive a monthly payment that consists of a *combination* of principal repayment along with interest on the mortgage pool that makes up the MBS.

What's the most common form of MBS?

Passthroughs that are issued by government sponsored enterprises (GSEs) such as Fannie Mae, Freddie Mac, and Ginnie Mae. The issuer of a passthrough will take a Servicing Fee and an Agency Fee, which will be deducted from the payments that are passed through to investors (so investors don't technically get all payments entirely passed through to them).

What's the major difference between a MBS and a traditional bond?

With a traditional bond, there will be semi-annual coupon payments with the principal amount paid back in full at a defined time. With MBS you get a combined principal and interest payment *monthly* – as mortgages are paid monthly – and there can be



prepayment of principal by mortgage owners (if they have more cash one month they may pay more down on their mortgage).

This creates timing uncertainty for the cashflows coming in, which is distinct from a corporate bond where cashflow timing and amounts are certain (unless they default!). Further, mortgage holders, if rates have decreased, can completely refinance their mortgage, which takes one mortgage out of the pool that makes up the MBS. Or mortgage holders may sell their home to move to a new one, thus paying off their mortgage early. MBS holders must be willing to accept this cashflow timing uncertainty.

Another way to say this, perhaps more simply, is that if rates decline all of a sudden you may have expected \$X in cashflows, but you'll get $\$X \times 1.2$ or something because more people are paying off their mortgage earlier than anyone anticipated.

What are the two major risk categories for MBS?

Prepayment risk, which is when a homeowner pays off a larger than normal part of their mortgage, pays off their entire mortgage due to selling their home, or refinances their mortgage due to rates declining.

Extension risk, which is when rates rise and people are less apt to pay down extra on their mortgage (prepayments decline) or refinance their mortgage (because it would be at a higher rate).

Thus, the two major risks surround cashflow uncertainty; you may end up getting more cashflows being disbursed than you anticipated, or you may get less (since you always assume some level of prepayment in your models).

This is probably the "hardest" question you'd ever get in an interview around MBS trading. As long as you show you have an intuitive understanding of the fundamental risks of the product, you will be *highly* impressive.

Basic Rates Q&A

What are treasuries? Can you categorize them?

Government debt issued by the U.S. Treasury that are backed by the full faith and credit of the United States.

There are several distinct categories based on maturity.

- Bills have a maturity of under a year and are traded by the money market desk
- FRNs are 2yrs and much less common than any other form of government debt
 - Began in 2014, floating rate linked to 3-month T-Bills with coupons paid quarterly



- Notes are 2, 3, 5, 7, and 10 years in maturity
- Bonds are 30 years in maturity

What are TIPS?

Most countries – including the United States – have inflation-protected securities that offer an interest rate over and above whatever the CPI happens to be. This locks in a certain real rate of return, whereas traditional Treasuries do not. TIPS are issued as 5 or 10 year notes or 30 year bonds.

What are treasury futures?

Futures allow market participants to take views on future rate movements in an off-balance sheet capacity. There are six futures:

- TU (2yr)
- FV (5yr)
- TY (10yr)
- TN (10yr, ultra-long)
- US (30yr)
- AUL (30yr, ultra-long)

Treasury futures are used routinely in hedging and are incredibly liquid just like the benchmark treasuries are.

Ultimately, a future obliges the seller to deliver an underlying security (not a future) sometime in the future (so a 10yr future would require a delivery of a 10yr Note).

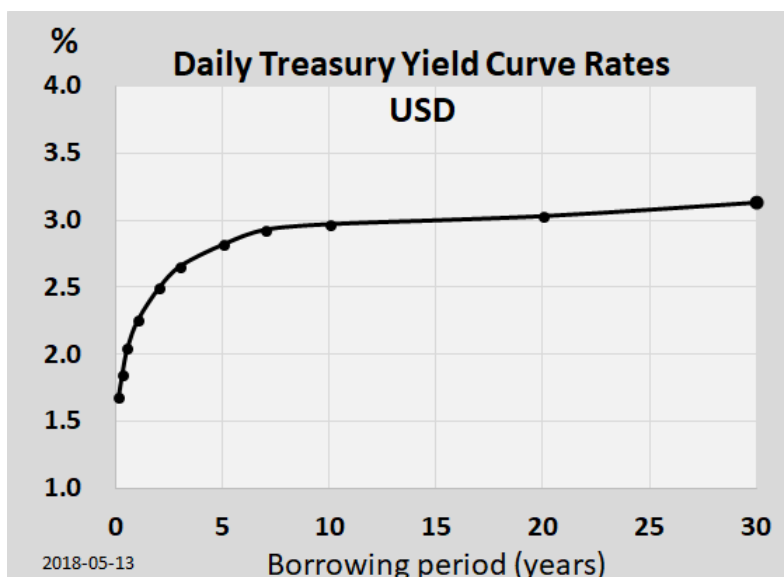
We've mentioned before the existence of a yield curve, what is that?

A yield curve is an interpolated curve showing the relationship between yield and maturity. The curve is interpolated because in reality there are not bonds at every single possible maturity point (e.g. maturities of exactly 892 days away, for example).

Instead, the curve has real points correlating to the maturities of the major bills, notes, and bonds and the line is essentially “filled in” between them. This is a rough description as yield curves interpolation is a bit more complicated.

Every country will have its own yield curve. You can see the U.S. Treasury's yield curve points here: <https://www.treasury.gov/resource-center/data-chart-center/interest-rates/pages/textview.aspx?data=yield>

And an illustration of the yield curve below:



What shapes do yield curves normally take?

Yield curves normally take an “upward sloping” shape indicating that as maturities increase so too do yields. This should make sense as investors should want more compensation (more yield) for holding government debt longer.

What do we mean by curve trades?

When we talk about a rates trader’s book being properly positioned and taking advantage of the market, this obviously doesn’t mean that if 2yr bonds are overvalued he or she just won’t have any!

Rates traders have such large books that they will always have millions of dollars’ worth of bonds that they may or may not like in order to facilitate client flow and generally be in the market.

Instead the way that a rates trader positions their book is by taking advantage of changes in the yield curve. For example, there’s nothing you can do about owning a lot of 2s. However, maybe you think that 2s will go down more than 5s. This would create a flattening of the yield curve between those two points in the yield curve.

Curve trades involve taking advantage of the relative changes in one bond compared to another along the yield curve by using treasuries and futures to be relatively overweight or underweight certain areas of the curve.

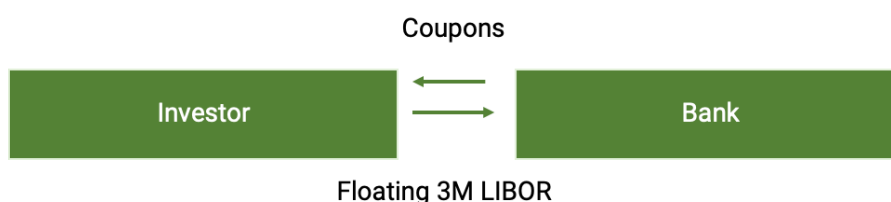


What routine news events have a large impact on rates?

- Federal Open Market Committee meetings, which occur eight times a year at routine intervals
- Speeches from Federal Reserve board of governors and the Fed chair
- Non-Farm Payrolls – monthly stat detailing the total number of US workers, which is released on the first Friday of every month
- Month / Year End – Quarter and year ends tend to involve rebalancing of portfolios. In particular from large institutional clients like endowments, pension funds, etc.
- Quarter End – US futures expire quarterly so market participants have to roll their holdings forward
- Auction results can lead to market turbulence if there is less than anticipated demand with yields coming in higher than expected (or vice versa)

What are traditional swaps?

There are many forms of swaps, the most traditional being the fixed for floating swap. The utility of this kind of swap is that if you think, for example, that rates will rise instead of receiving fixed coupon payments you can get a floating rate that will increase as rates do (thus making you better off in a rising rates environment, worse off in a declining rates environment)



What characterizes swaps on day one?

Each leg or side of the swap (the fixed side or the floating side) will have equal value at initiation. In other words, it's a fair trade at the starting point and one side will "win" if what drives the value of the swap moves in their direction (e.g. rising rates will help those who receive floating interest and hurt those that receive fixed interest).

Backing up for a second, what is LIBOR (that the floating leg of an IRS is based on)?

So, a fixed rate payer pays a fixed coupon amount and at the initiation of a swap an equivalent amount of 3M LIBOR (plus a spread) is paid as the floating leg.



The London InterBank Offered Rate (LIBOR) is the most widely used benchmark in finance. It is the interest rate at which banks borrow funds from other banks at large size in the London InterBank market. It is updated daily when a number of banks submit what they can borrow at.

LIBOR underpins nearly every floating piece of debt; from conventional loans to swaps. You'll often see this quoted as L + 300bps or something, which just means the LIBOR rate at a given time plus 3% (300 basis points).

What is important about the future of LIBOR?

What's important to know is that LIBOR will be phased out by 2022! This is due to the LIBOR manipulation scandal that saw some banks collude to artificially increase LIBOR rates by submitting artificially high bids. LIBOR will largely be replaced by SOFR; you can read Morgan Stanley's report on it here: <https://www.morganstanley.com/ideas/libor-its-end-transition-to-sofr>

Basic FX Q&A

What is a spot FX trade fundamentally?

A spot FX trade is one of the simplest in all of S&T. It is simply an agreement between two parties to exchange a specific amount of two currencies at the "FX rate".

What is the FX rate?

The FX rate is the amount of the second currency exchanged per unit of the first. USD/CAD is 1.31 at present; meaning 1.31 Canadian dollars can be exchanged per (one) unit of U.S. dollars.

What are the three major forms of FX trades?

- Spot trades: trades at the present FX rate that are settled in two business days
- Forward trades: trades that will be settled in some amount of time greater than two business days
- Swap trades: a trade made at time X that will be reversed at time X + n (where n is some amount of business days in the future)

What are the most common FX trades (by total dollar volume)?

Roughly 40% occur in the spot and swap market each. 10-15% in the forward market and 5-10% in the FX options market.



What are currency pairs?

Currency pairs simply refer to the two legs of a FX trade; for example, EUR/USD, CAD/USD, USD/JPY, etc.

What affects FX rates?

The right answer is probably everything, which is why many FX traders who begin on the sell-side (within S&T) will move to global macro funds.

It's important to recognize that in FX everything is relative; a currency is only "strong" or "weak" in relation to the currencies it trades against. So, what drives strength or weakness will be relative differences in monetary policy (interest rates and central banking policy), economic statistics (inflation, economic growth, etc.), and politics (political stability, foreign policy stances, domestic spending objectives, etc.).

What is a pip?

Pips depend on the currency being looked at, however for USD it is \$0.0001. So, in our prior example we said the spread was 60-65 pips or \$0.0060 / \$0.0065.

What are FX call and put options?

FX call and put options make up a reasonably small share of the overall FX market, however there will often be traders who specifically deal with FX options on a desk (perhaps with overlapping swap trading responsibilities).

FX call and put options are no different than what you would expect. Buying a call option gives one the right, but not the obligation, to buy an agreed upon amount of currency at an agreed upon strike price. Buying a put option gives one the right, but not the obligation, to sell an agreed upon amount of currency at an agreed upon strike price.

The strike price for a call will, of course, be above the spot price (so out-of-the-money, or OTM, at initiation). If the currency moves in the direction of the strike price the call will become more valuable.

Basic CLO Q&A

Philosophically, what is a CLO?

A CLO is the largest, most actively traded form of securitized product that exists today. CLOs operate by taking leveraged bank loans, pooling them together, and then creating tranches of securities (bonds) out of these that are backed by the cashflows of said loans.

The point of a CLO fundamentally is risk transformation. You take loans that are individually risky, pool them together, and then give first priority to the highest class of



bonds (the highest tranche) in the CLO. This makes this top tranche very safe and thus they will get a AAA rating from a credit rating agency.

Of course, not every tranche will be as safe! Lower tranches will have a lower payment priority (only getting cashflows after the other tranches have been paid) and so have higher levels of risk and reward (in the form of the coupon payment tied to them).

The important point once again is that CLOs provide risk transformation; turning uniformly risky loans into bonds that are either very safe, or quite risky themselves.

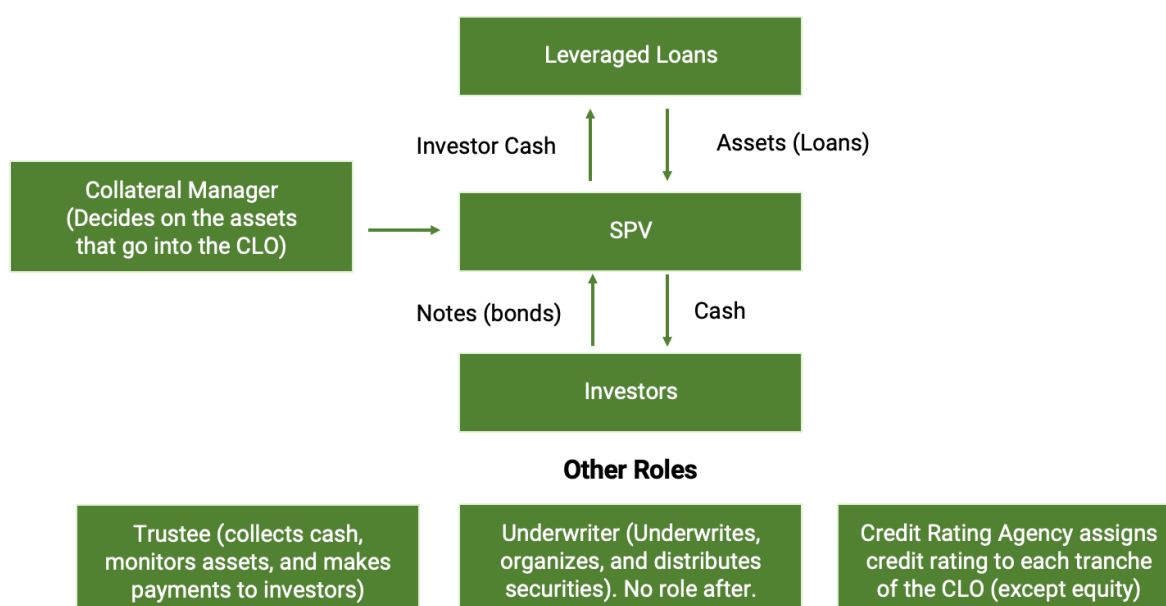
Who are the parties involved in a CLO at issuance?

CLOs will always have a SPV (special purpose vehicle) that can be thought of as the nerve center via which all counterparties interact.

Investors will put cash into the SPV in return for notes (bonds) at the various tranches. A collateral manager will then be responsible for filling the SPV with leveraged loans that will make up the CLO (these can either be new issuance or loans already trading in the secondary market).

When the CLO is filled with levered loans, a credit rating agency will assign credit ratings to each tranche of debt (outside the “equity” tranche). The underwriter is responsible only for underwriting the notes (bonds) and distributing them to investors; they have no role after this.

The trustee will then collect the cashflows from the loans and send them to investors in order of their priority (as specified in the notes).



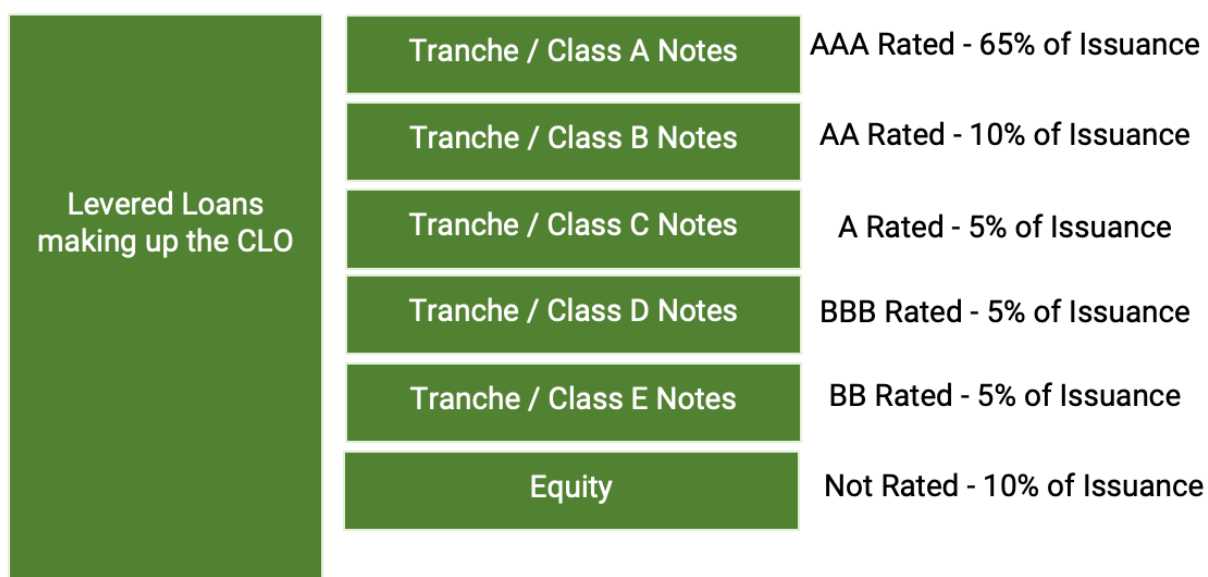


Note: The arrows between SPV / Investors should obviously be reversed here. I lost the source file for my little graphic so will re-create this when I have time. Obviously investors put cash into the SPV and get Notes (bonds) in return (not the other way around).

How is a CLO itself structured?

As mentioned, the CLO itself will be broken down into several tranches. These tranches will have decreasing levels of priority on the underlying cash flows of the levered loans that make up the CLO (to be discussed later).

The top tranche, which makes up the *majority* of the notes issued by the CLO, will often be structured to ensure that it gets a AAA rating from credit agencies (thus making investors feel it is safe). The bottom tranche, called “equity”, will have the lowest payment priority and is not rated and viewed as being quite risky.



An important point in the graphic above is that the levered loans that make up the CLO often *overcollateralize* the issuance. What this means is that in theory, if none of the loans default, then equity should get much more than what they put in!

However, the risk equity runs is that levered loans are risky and some that make up the CLO may end up defaulting. So even though the CLO itself is overcollateralized at issuance, equity and some of the tranches above it may be at risk of not being paid back in full over time.



If you're any of these buyers, why not just buy pieces of loans that are actively traded in the secondary market?

Many large loans will be syndicated after issuance and traded on the secondary market. This raises the question of why one would not just buy the exact loans they like directly, as opposed to CLOs that combine so many levered loans together.

There are two primary answers. First, as already discussed, a CLO provides you safety through the payment priority the tranche of debt you're buying has. Second, CLOs can benefit from the law of large numbers; instead of carefully studying one levered loan and saying you think it has a low chance of default, you can buy a small piece of many different levered loans indirectly via a CLO and thus have less specific credit exposure. CLOs offer a form of diversity; giving you general credit exposure to levered loans instead of specific credit risk to just a few names.

When do CLOs end? They can't go on indefinitely?

CLOs wind down naturally as the levered loans that make them up mature. This can take up to 10 years and this final phase of a CLO's life is called the wind down period. Of course, CLOs can't go on indefinitely as all levered loans eventually mature (usually in 7-10 years at most) or default.

Will a CLO desk hold all tranches of a certain CLO?

Not necessarily! A CLO desk's inventory can involve holding notes in just a few of the tranches of a certain CLO. Sometimes certain CLOs are dominated by market participants who are interested in holding the notes for the long-term and thus they are not actively traded in the secondary market.

Basic Muni Q&A

Simply put, what are muni bonds and who issues them?

Municipal bonds are bonds issued to finance capital expenditures that are in the public interest or represent a public good. These capital expenditures can include things like highways, wastewater plants, electrical grids, schools, etc.

Municipal bonds are issued by over 100,000 unique entities in the United States including: highway authorities, power authorities, hospitals, school districts, and all levels of government below the federal government (states, cities, counties, etc.).



Explain the tax exemption part of muni bonds to me?

Any discussion of the tax except nature of muni bonds should begin by you saying it's complicated! The nature of tax exception will differ depending on locality and type of issuer and these things change over time slightly.

Broadly speaking, when we talk about tax exemption with muni bonds we mean that interest income earned from most municipal bonds is not taxed at the federal level.

So for example, if you own \$1,000,000 in face value of a muni bond paying 2% then you get \$20,000 in annual income free and clear of tax. This makes muni bonds appealing to high net worth individuals looking for relatively safe, tax free income.

What are the two kinds of muni bonds that exist?

Generally speaking, you can break down muni bonds into either being general obligation bonds (GO bonds) or revenue bonds. As the names imply, GO bonds are backed generally by the taxing power of the state whereas revenue bonds are tied directly to a certain project.

What have the recent trends in muni bond issuance been?

If you bring up in an interview that you're interested in municipal bonds, then the assumption will be that you follow the market a bit. No one is expecting you to be an expert in such an esoteric product, but you should generally have an idea if the market is hot or not.

Right now – in 2020 / 2021 – the municipal bond market is trending upwards: <https://www.lordabbett.com/en/perspectives/fixedincomeinsights/behind-surge-taxable-muni-bond-issuance.html>

The rise in muni bond supply this year is likely due to both the increased funding needs that have arisen due to COVID-19 and the increased demand for muni bonds due to the record low yields in treasuries (that are likely to stay for a significant amount of time as can be seen in both futures and out the yield curve).

What's the liquidity of munis like?

This largely depends on the issuance size, time since issuance, and any market developments that have occurred.

This is a good interview question, because there's no quick answer. The reality is that if you have a large issuance, from a large issuer, the bond will stay liquid for years potentially and be actively traded by pension funds, hedge funds, and mutual funds.



However, for small insurances the market may be liquid for a few months, but will dry up as clients who buy munis do tend to hold them for the longer term rather than trade them back and forth.

How do you think about the actual yield of a muni bond?

The major differentiator of a muni bond is that it is (at least) tax exempt at the federal level. Therefore, when talking about the yields of XYZ muni bond and say an Apple corporate bond it's not comparing – pardon the pun – apples to apples.

Instead, we need to adjust the muni yield quoted to reflect the tax savings. So for example, if we have a muni bond yielding 2.0% and the marginal tax rate is 30% then the Tax Adjusted Yield of the muni bond will be: (Tax Adjusted Yield) = (Quoted Yield)/ (1-tax) or $2.0\% / 1-30\% = 2.86\%$.

So the fact that this muni bond is tax exempt means that the actual yield, after adjusting for taxes, is 86 basis points (0.86%) higher than it is quoted.

Why are you interested in muni bonds?

An answer to this common interview question is:

I find muni bonds interesting because there are so many issuers – around 100,000 – all issuing unique bonds to fund unique, real-world projects. This makes the market obviously quite illiquid and provides a challenge for those in S&T – whether in trading or in sales – to be fully up on the market; understanding the trends between, for example, housing and industrial development munis; and trying to make good markets for clients while not getting caught on the wrong side of any given trade.

It's also a very opaque market where I think you can learn a lot by being "in the mix" compared to other products where information is more readily disseminated to the public and market professionals alike. I find the niche aspect of munis to be a real pro, whereas others may view that as being a con.

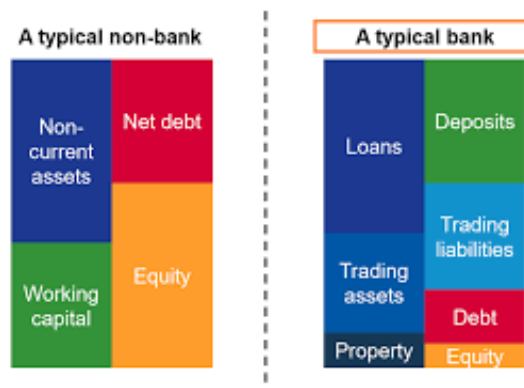
Basic Regulation, Economics, and Corporate Finance Q&A

How does a bank balance sheet compare to that of a traditional company?

The primary difference is that the assets of a bank will be made up of the securities they hold; either primary issuance (loans) or bought on the secondary market (such as securities bought for trading purposes).



On the liabilities side there will be a mix of equity, debt, and deposits. Deposits refer to retail deposits made by consumers (e.g. if I put \$100,000 in my savings account, that is a liability on the banks' balance sheet).



What does marked-to-market refer to?

Lots of regulations around banks surround marked-to-market accounting. Simply put, this is where every day certain securities held by the bank (assets) are valued at their fair market value (what they are trading at in the market).

As you can imagine, during volatile periods this can create the appearance of stress. For example, if you have credit assets fall in value across the board by 5% - due to some large negative event that caused the market to crash broadly - all of a sudden your "assets" side of the balance sheet has shrunk quite a bit and the bank could appear stressed (because liabilities would stay roughly the same).

This was a large issue in the financial crisis. Mortgage backed securities fell significantly in value - even though many viewed them as being much safer securities over the long-run than the market was pricing - but nevertheless banks were forced to sell them off at cheap prices in order to stay "solvent".

How do banks make money?

This is actually sometimes asked in interviews and is a good question. It shows that you understand (roughly!) how a bank works. For a retail bank - one that actively takes deposits and lends to consumers - with an investment banking side the following streams exist. This would include banks you'll likely apply to like J.P. Morgan, BAML, and even now Goldman Sachs (since they are now taking retail deposits via Marcus).

- Trading / Investing profits
 - Profits from issuance of securities and from the trading of securities in the secondary market
 - This can include proprietary investments (where possible) and investments in credit or PE funds started by the firm itself
- Fees from Services



- In investment banking (M&A, ECM, DCM) and asset management, banks earn a fee from doing an activity, rather than warehousing risk like in trading
- For ECM and DCM there is some element of warehousing risk, however it is distinct from how risk is warehoused by traders
- Net interest income
 - The difference between what you are paying depositors and what you are earning on money lent out
 - Inclusive of this are interest payments earned from securities held by traders

You should briefly take a spin through the 10-K or 10-Q documents of the bank you're interviewing at to get an idea of the mix of income they have. Every bank will be different. For example, Goldman earns little net interest income and roughly equal amounts from trading and investment banking fees.

JPM and BAML on the other hand are incredibly large banks – with lots of retail customers – who will earn lots of net interest income (in conjunction with trading profits and investment banking fees). This makes them more sensitive to lower rates as less net interest income is earned all things being equal.

What are the two major regulatory changes that have affected S&T over the last decade?

The two major regulatory changes are the full implementation of Basel 3 (2019) and Dodd-Frank (came into effect in 2015-2016). Note that both of these were long forecast by regulators and slowly introduced.

While both regulations are very complex and lengthy, the best way to boil them down is that Basel 3 is focused in on capital controls (keeping banks sufficiently liquid, in their view) and Dodd-Frank is focused in on trading controls (keeping banks from doing risky things).

Dodd-Frank is a U.S. based law with the most relevant portion – for S&T – being in the form of the Volcker rule, which lays out rules surrounding taking proprietary positions as a trader. These positions are defined as those that are not taken in order to facilitate client transactions, but rather purely to speculate.

Here's a good flow chart on Dodd-Frank:

https://www.davispolk.com/files/DavisPolk_Final_Volcker_Rule_Flowcharts_Prop_Trading.pdf

Basel 3 comes out of the Bank for International Settlements (BIS) and has had a number of iterations going back to the 1990s.



What are two Fed reports that are useful to understand Fed policy and how the economy is doing?

The Monetary Policy Report, which is used to brief Congressional members as to the state of the economy: https://www.federalreserve.gov/monetarypolicy/mpr_default.htm

The Beige Book is a collection of economic indicators and anecdotes collected by Federal Reserve governors across the country:
https://www.federalreserve.gov/monetarypolicy/mpr_default.htm

What's GDP and its components?

Gross domestic product measures the goods and services created within an economy during a given time.

The equation – via the expenditure approach – is $C + I + G + NX$.

- C = Consumer
 - Personal Consumption Expenditure
 - Durables
 - Non-Durables
 - Services
- I = investment
 - Investment and change of inventories
- G = Government
 - Federal spending
 - State and local spending
- Net Exports
 - Net exports of both goods and services

The U.S. economy – like most major economies – is driven primarily (around 65-75%) by consumer spending, followed in roughly equal measure by investment and government depending on the year. Net exports can often be negative for developed economics.

What are the two primary measures of inflation?

CPI, which is the change in price (quarter-over-quarter or year-over-year) of a basket of goods and services that a “standard individual” would buy in a given year.

Core CPI is this same basket, stripped of food and energy given the historical volatility of those.



What is a market measure of inflation commonly utilized?

The 5Y5Y swap rate, which is an actively traded swap of a five-year maturity that doesn't begin until five years from now. This swap rate indicates what the market expects inflation to be over a five-year period beginning in five years. Usually it is around 2% given that 2% is the Fed target for inflation.

What are the two major US employment numbers that market participants will watch?

- Initial jobless claims
 - Every Thursday at 8:30EST, which shows the number of people claiming unemployment over the previous week
- Non-Farm Payrolls
 - First Friday of every month at 8:30EST, which shows the change month-over-month of the number of people employed
 - This is an employment measure, while initial jobless claims is an unemployment measure

What is PMI?

PMI stands for the purchasing managers index and is conducted in a number of countries / regions monthly. It asks purchasing managers whether conditions (including output, forecasts, etc.) have improved or not. If 100% of respondents say improvements are being seen then the PMI would be 100. If 0% of respondents say there are improvements, the PMI would be 0.

If only 50% of respondents say conditions are improving, the PMI would be 50. Generally, whenever PMI dips below 50 that raises alarm bells as it is indicating the majority of respondents are not seeing improving conditions relative to prior periods.

What are the future value and present value formulas?

$FV = PV(1+r)^n$ where r is the interest rate and n denotes the number of periods.

PV is just the rearranged formula, or: $PV = FV/(1+r)^n$. Note that the denominator of the PV formula operates as a discounting mechanism; the more periods that go by (the higher the n value) the smaller the PV will end up being as the denominator will be growing.

What are the three financial statements?

- Income Statement
 - Takes into account information from the current period that affects taxes
- Balance Sheet



- The balance sheet represents a snapshot of the assets, liabilities, and shareholders equity of a company at a given time
- Cash Flow Statement
 - The CFS has three distinct sections: cashflow from operations (CFO), cashflow from investing (CFI), and cashflow from financing (CFF).
 - Unlike the income statement that will show non-cash expenses – like depreciation – the cash flow statement purely shows what affects the underlying cash balance of the company

What is EBITDA? Why is it important?

EBITDA (earnings before interest, tax, depreciation, and amortization) is a common multiple used in company valuation and in the covenants that govern debt documents.

EBITDA can be found by taking net income then adding back interest expense, tax expenses, depreciation, and amortization. EBITDA is a normalizing metric that strips the company of capital-structure or company-specific choices in expensing and thus allows for better comparison between companies.

What are the three primary ways to value a company?

In M&A banking – as you may know – there are three primary valuation techniques that will be utilized to narrow in on an appropriate valuation range for a company: precedent transaction analysis, comparables analysis, and discounted cash flow analysis. A LBO analysis may be done as well to set a floor on the valuation of a company (if a sponsor – meaning private equity fund – were to buy it).

A very brief overview of each is as follows. These are good to know, but you shouldn't expect traditional investment banking technicals in your S&T interview.

- Precedent Transaction Analysis
 - Looking at past M&A transactions and applying the purchase EV/EBITDA multiple to the company you are looking at
- Comparables Analysis
 - Looking at current publicly traded companies that are comparable to the one you're looking at, finding the EV/EBITDA multiple of each, averaging them out, and applying the multiple to the company you're looking at
- Discounted Cash Flow Analysis
 - Forecast the FCF of the company for five years and discount it back to today, then apply a terminal multiple to year five FCF and discount that back as well. The summation of these discounted cashflows and terminal value will yield an enterprise value (what the company is hypothetically worth)
- LBO Analysis



- Complete a LBO model that shows the maximum amount that a private equity fund would likely pay for the company; private equity funds will pay a lower EV/EBITDA than traditional buyers so this sets a floor on valuation

What is enterprise value conceptually?

Enterprise value is the value of the company if one were to buy it outright. While EVs can get quite involved; the most basic formula is $EV = \text{equity value} + \text{preferred stock} + \text{outstanding debt} + \text{minority interest} - \text{cash and cash equivalents}$

What's the difference between enterprise value and equity value?

Equity value is simply the market value of the company; the amount of fully diluted shares outstanding times the share price.

When we talk about valuation multiples for a company, what are we almost always referring to?

Enterprise Value (EV) / EBITDA. Sometimes EV/revenue or EV/EBIT will be utilized. However, EBITDA is almost always the norm as it eliminates D&A variability across companies. If for some reason a company has a distorted EBITDA – perhaps because they are a young, fast growing tech company – a different multiple like EV/revenue or EV/users will be utilized.

How do we arrive at free cash flow utilized in a DCF?

In a DCF unlevered FCF is utilized, not levered FCF (levered takes into account debt payments).

FCF will simply be:

- EBIT (1-tax rate)
- + Non-Cash Expenses like depreciation and amortization
- - changes in working capital
- - capex (capital expenditures)
- = Unlevered FCF

What is the primary difference between a DCF and the other major valuation techniques of precedent transactions and comparables analysis?

A DCF projects forward free cash flows – which are derived from projections of financial statement line items – so is untethered to current market prices or past transactions. Precedent transactions and comparables are tied to the market, however, as they involve either past transactions (precedent transactions) or equity value in the enterprise value formula (in the case of comparables analysis).



Basic Money Markets Q&A

Philosophically, what are money market securities?

Money market securities are a broad set of securities that are usually traded by a singular desk (the money markets desk). These securities have one thing in common: duration. The duration (or maturities) of money market securities are less than a year.

Money market securities can be secured or unsecured and can be government-backed – such as Treasury bills – or issued by corporations.

Broadly speaking, what are the two kinds of categories of money market securities?

Those that have interest tied to them and those that don't. For example, certificate of deposits, Fed Funds, and repos all are issued near or at par and have an interest payment tied to them like you are used to seeing with Treasury bonds or corporate bonds.

However, because of the short duration of money market securities there are also discounted securities that have no interest payments associated with them. Instead, these securities are issued *below* par, but mature at par. So, your effective gain on the security is the difference between where it was issued (e.g. 99.75) and where it matured (100).

What are some examples of money market securities?

As mentioned, money market securities have different names than traditional financial securities. The main money market securities are: Repos, Fed Funds, Treasury Bills, Commercial Paper (CP), and Certificates of Deposit (CDs).

What are repos?

Repos are one of the most fascinating areas of money markets; generally, there will be the most people on a money markets desk focused on repos than any other money markets type of security.

Further, unlike the other money market securities that generally show very little volatility – except in extreme financial conditions like during 2007-09 – repos have significant flares of volatility that can lead to large losses.

<https://www.bloomberg.com/news/articles/2020-01-06/why-the-u-s-repo-market-blew-up-and-how-to-fix-it-quicktake>



Repos operate under a simple concept: an individual who owns the security enters into a contract – a repo – whereby they sell the security and agree to buy it back at a specified later date at an agreed to price.

Basic Equity Derivatives Q&A

What's the real price of option?

Intrinsic value + time value.

What assumption does Black-Scholes make that are wrong?

- Normal distribution
 - Pronounced skew or negative tail in actual returns on shares, meaning that there is a bigger chance of significant losses than is built into the shape of the bell curve
- Continuous random walk
 - Assumes that returns follow random walk, which is not what is actually observed
- Dynamic hedging
 - Assumes it is possible to delta hedge without transaction costs and without liquidity constraints
- Fixed vol
 - Assumes you know the level of vol which stays constant over the life of the option

What are the greeks really?

Partial derivatives of the option pricing model. Delta, theta, vega, and rho. This means that they measure the change in the calculated option value for a given change in one of the inputs to the model, all the other inputs remaining constant.

What is delta?

Measures the change in the option value for a small change in the price of the underlying asset, assuming all other inputs to the model are held constant. Long call and short put are delta positive (price of the underlying rises, delta is positive). Short call and long put are delta negative if the position loses value the underlying increases.

What is gamma?

Measures the changes in option delta. Useful indicator of how stable or otherwise a delta hedge is likely to be. Graphically, gamma is a measure of the curvature (or convexity) in the relationship between the value of an option and the spot of the underlying. The greater



the curvature the more rapidly the delta (the slope or tangent on the option pricing curve) will change. Essentially tells trader to buy or sell more shares to properly hedge delta.

What is theta?

Change in the value of an option as time elapses (with all other inputs to the pricing model remaining the same). Really measures rate of decay in the time value of an option.

What's vega?

AKA kappa. The value of an option is also sensitive to the assumption made about the vol of the underlying. This sensitivity is measured by vega. Vega measures the change in the option for typically a 1% change in the assumed vol of the underlying.

What does vega of \$0.2 mean?

Means for 1% change in vol the option will increase or decrease in value by 20 cents. This is on a per share basis.

What is Rho?

Change in an option's value for a given change in interest rates.

Time value highest when strike price is similar to?

Spot. Time value is highest because you're right around the strike price so very sensitive to movements above or below it.

When is gamma low?

When OTM or ITM. Tend to be most sensitive when ATM.

What's implied vol?

Vol assumptions built into the actual dollar price of an option. Obtained by operating the pricing model backwards. Rather than using vol assumption to determine value of an option, use the dollar price at which it is trading in the market in order to determine the vol level that would generate the observed price.

What's the VIX?

Vol index was introduced in 1993. Measures vol expectations built into S&P index options with 30 days to expiry. Derives market expectations of vol on S&P directly, rather than extracting or backing out an implied vol assumption using an option pricing model. VIX is calculated in real time throughout the trading day.



What's a convertible bond?

Owner has the right to convert the bond into predetermined number of shares. Number of shares it can be converted into is called the conversion ratio. The current value of those shares is called the parity or conversion value of the CB.

What do convertible bonds (CBs) have embedded in them?

A call option. Can't be split off and traded separately from convertible bond. Only exercised through conversion.

What is a long forward position?

Agree to buy share in a year's time at fixed price (say \$100). Short forward position if agree to sell the share to someone at fixed price of \$100 in a year.

What's attraction of forward starting products?

You get vega without gamma (for short term vol). No paying of theta.